

Audio Preamplifier

Design, Construction and Implementation

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Document Type

Independent Engineering Project Report

Project completed: June 2022

Abstract

This report documents the design, construction, and implementation of a standalone stereo analog audio preamplifier completed in June 2022. The system combines conventional line-level source selection, a dedicated moving-magnet phono input with RIAA equalisation, line-level voltage gain, a logarithmic stereo volume control, three parallel stereo line outputs, and a separately controlled low-pass output for a subwoofer system.

The principal line preamplifier is based on Rod Elliott’s Project 88, while the phono section is based on Project 06 and the regulated symmetrical power supply follows the topology of Project 05 [3, 2, 1]. A separate low-pass-filter PCB is based on Project 155 [4]. The final implementation uses LME49720 dual audio operational amplifiers, a nominal $\pm 15\text{ V}$ regulated supply, local supply decoupling, and a controlled star-ground arrangement.

The main PCB schematic was developed in Autodesk Eagle. The schematic was subsequently transferred to EasyEDA, where the two-layer PCB layout was completed and prepared for manufacture by JLCPCB. The fabricated main PCB uses 1.6 mm FR-4 material and 1 oz copper. The completed electronics were installed in a metal enclosure together with a $15 - 0 - 15\text{ VAC}$ toroidal transformer, front-panel controls, rear-panel RCA connectors, and internal signal and power wiring.

The report places particular emphasis on practical implementation issues encountered during construction. Audible input crosstalk and mains-related hum were reduced by replacing susceptible signal wiring with coaxial cable, improving cable routing, electrically isolating the RCA connectors from the metal enclosure, and bringing the circuit returns and protective-earth connection to a dedicated chassis star-ground point. No formal laboratory characterisation of gain, distortion, frequency response, or noise was performed; numerical values presented in this report are therefore either design values, component values, or theoretical calculations rather than measured performance data.

The completed project demonstrates the relationship between analog circuit design and its physical implementation, including power-supply design, component selection, PCB layout, grounding, shielding, interconnection, mechanical construction, and practical

troubleshooting.

Acknowledgements

I would like to thank everyone who provided support, encouragement, and technical discussion during the development and construction of this project.

In particular, I would like to thank Chevas Lukas for his valuable help, support, and practical assistance during the development of the preamplifier.

I am also grateful to Rod Elliott for making the Elliott Sound Products reference designs and technical articles publicly available. These resources provided the basis for several circuits incorporated into the project and were valuable during the design and implementation process.

Finally, I would like to thank my family and friends for their support and encouragement throughout the construction of the project.

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Abbreviations

AC	Alternating current
DC	Direct current
EDA	Electronic design automation
FR-4	Glass-reinforced epoxy laminate used for PCB construction
MM	Moving-magnet phono cartridge
PCB	Printed circuit board
RIAA	Recording Industry Association of America equalisation characteristic
RMS	Root mean square
VA	Volt-ampere

Chapter 1

Introduction

1.1 Background and Motivation

An audio preamplifier forms the interface between source equipment and a power amplifier or another line-level destination. Although the power handled by a preamplifier is small, the circuit must process low-level signals without introducing unnecessary noise, interference, instability, or signal loss. The practical quality of such a system therefore depends not only on the selected circuit topology but also on the power supply, component selection, grounding arrangement, PCB layout, enclosure, and internal wiring.

The motivation for this project was to construct a complete standalone analog preamplifier rather than a single experimental circuit. The unit was intended to combine multiple source inputs, a dedicated phono stage, a line-level preamplifier, a user volume control, several outputs, and a separate subwoofer-filter function in one metal enclosure.

The project was designed and physically completed in June 2022. The present report was prepared later in order to document the engineering process, reference circuits, implementation decisions, construction details, and practical lessons learned from the finished hardware.

1.2 Project Objectives

The principal objectives of the project were to:

- construct a complete stereo analog preamplifier for use with an external power amplifier;
- provide two conventional line-level source inputs and one moving-magnet phono input;

- provide front-panel source selection and a stereo master-volume control;
- include a dedicated RIAA phono preamplifier for vinyl playback;
- provide three parallel stereo line-level outputs;
- include a separately adjustable low-pass output for a subwoofer application;
- use an internal regulated symmetrical power supply;
- implement the design on professionally manufactured PCBs;
- minimise audible hum and crosstalk through suitable grounding, shielding, and wiring; and
- integrate the complete electronic system into a mechanically practical metal enclosure.

1.3 System Requirements

The preamplifier was intended for conventional domestic analog audio use. The line-level signal path had to accept normal source equipment and provide a controlled output suitable for a separate power amplifier. The phono path had to provide the additional gain and equalisation required by a standard moving-magnet cartridge.

The electronic circuitry was required to operate from symmetrical regulated supply rails so that the analog signal could be processed around a 0 V reference. The final supply target was ± 15 V.

The enclosure had to contain the toroidal transformer, main PCB, subwoofer filter PCB, power switch, source selector, volume control, subwoofer controls, rear-panel RCA connectors, and all necessary internal wiring. Because the transformer and mains wiring are located in the same enclosure as low-level audio circuitry, physical separation, grounding, and shielding were important design considerations.

1.4 Scope and Methodology

The scope of this report includes circuit selection, adaptation of reference designs, power-supply implementation, schematic capture, PCB design, component selection, assembly, enclosure integration, grounding, wiring, troubleshooting, and final functional evaluation.

The report intentionally distinguishes between implemented hardware and reference-design features. For example, the Project 88 reference design includes a balance-control

arrangement, but the final preamplifier does not use a balance potentiometer. The balance section was bypassed in the constructed unit and is therefore not treated as an active user control in this report.

No calibrated laboratory characterisation was carried out on the completed unit. Consequently, the report does not claim measured values for total harmonic distortion, signal-to-noise ratio, channel separation, absolute frequency response, or exact supply-rail voltage. Where calculations are included, they are identified as theoretical design calculations.

1.5 Reference Designs

Four Elliott Sound Products circuits formed the principal technical basis of the project:

- **Project 05 – Power Supply for Preamplifiers:** used as the basis of the fixed $\pm 15\text{ V}$ regulated supply [1];
- **Project 06 – Hi-Fi Phono Preamp (RIAA Equalisation):** used for the moving-magnet phono input [2];
- **Project 88 – High Quality Audio Preamp:** used for the principal line-level preamplifier [3]; and
- **Project 155 – Variable High-Pass and Low-Pass Filters:** used for the adjustable low-pass subwoofer-filter section [4].

These reference circuits were incorporated into a single practical system and adapted to the available controls, enclosure, PCB layout, and desired input/output arrangement.

1.6 Organization of the Report

Chapter 2 presents the complete system architecture, regulated power supply, and grounding arrangement. Chapter 3 describes the input architecture, source selection, and moving-magnet phono preamplifier. Chapter 4 discusses the main line-level preamplifier, the implemented gain configuration, volume control, parallel outputs, subwoofer filter, and final operational-amplifier selection.

Chapter 5 documents the PCB-development workflow, board construction, component placement, internal wiring, and mechanical integration. Chapter 6 presents the practical troubleshooting and optimisation carried out after assembly. Chapter 7 summarises the final implementation, limitations, and possible future improvements. Complete schematics and additional project photographs are included in the appendices.

Chapter 2

System Architecture, Power Supply and Grounding

2.1 System Overview

The completed hardware is a standalone stereo analog preamplifier housed in a metal enclosure. The front panel provides the principal controls required for normal use: master volume, input selection, subwoofer level, subwoofer cut-off frequency, and mains power.



Figure 2.1: Front view of the completed standalone audio preamplifier.

The rear panel provides RCA connections for the available sources and outputs. Internally, the enclosure contains the main preamplifier PCB, the separate low-pass-filter PCB, the toroidal power transformer, the front-panel controls, and the associated signal and power wiring.

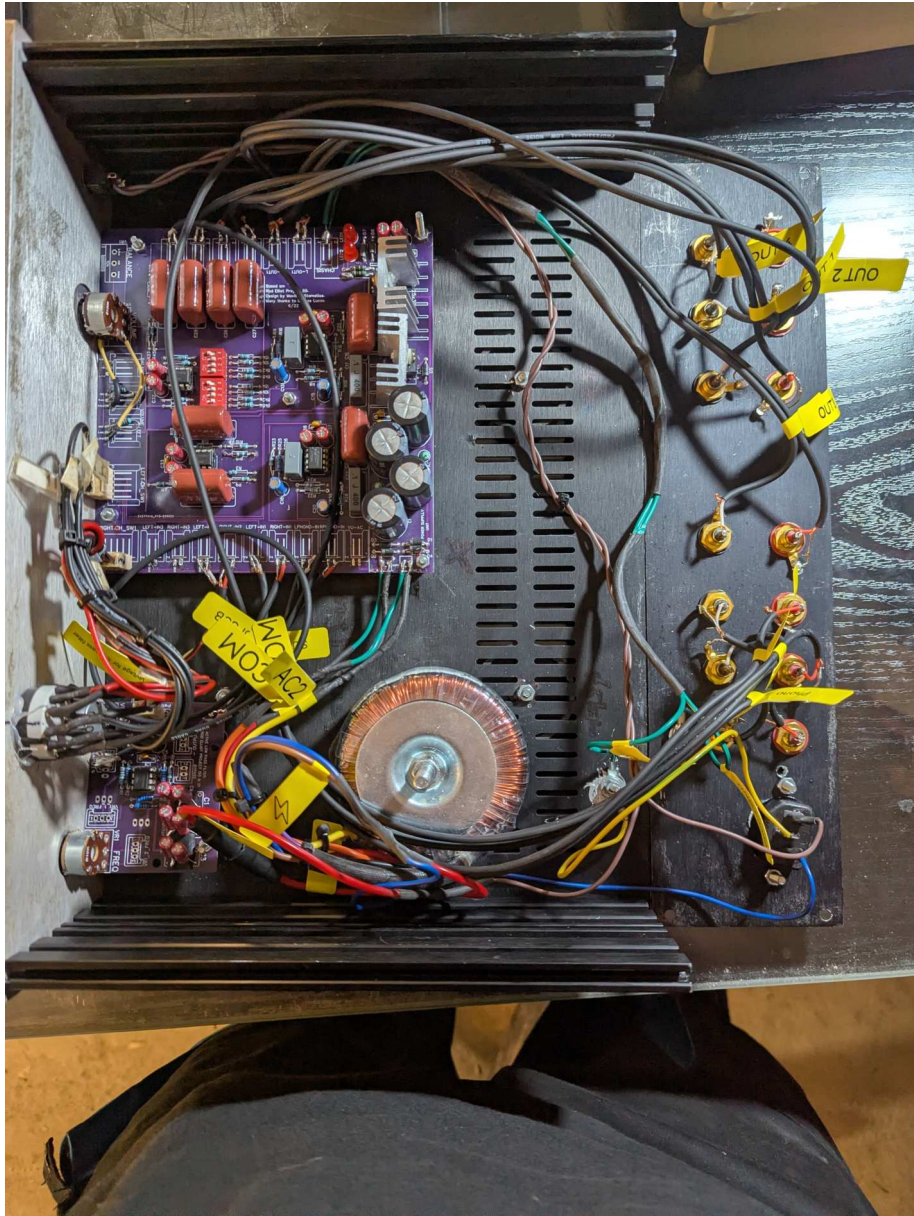


Figure 2.2: Top-down internal view of the completed preamplifier.

2.2 Functional Architecture

The principal audio path can be represented as

Source → Input Selection → Required Input Conditioning → Line Preamplifier → Volume Control →

For the two line-level inputs, the selected source is passed directly to the line-preamplifier section. For the phono input, the moving-magnet cartridge signal first passes through the dedicated Project 06 RIAA stage before entering the normal line-level signal path.

A separate low-pass-filter board provides the subwoofer-output function. It operates at line level and is independent from the three main stereo output pairs.

2.3 Power Supply

2.3.1 Supply Requirements and Topology

The LME49720 operational amplifiers used throughout the final implementation operate from a symmetrical supply. A nominal $\pm 15\text{ V}$ supply was selected because it provides useful analog headroom while remaining within the operating range of the device [5].

The power-supply topology follows the fixed-regulator form of Elliott Sound Products Project 05 [1]. The implemented supply consists of a centre-tapped transformer, discrete rectifier diodes, smoothing capacitors, positive and negative three-terminal regulators, and local supply bypassing.

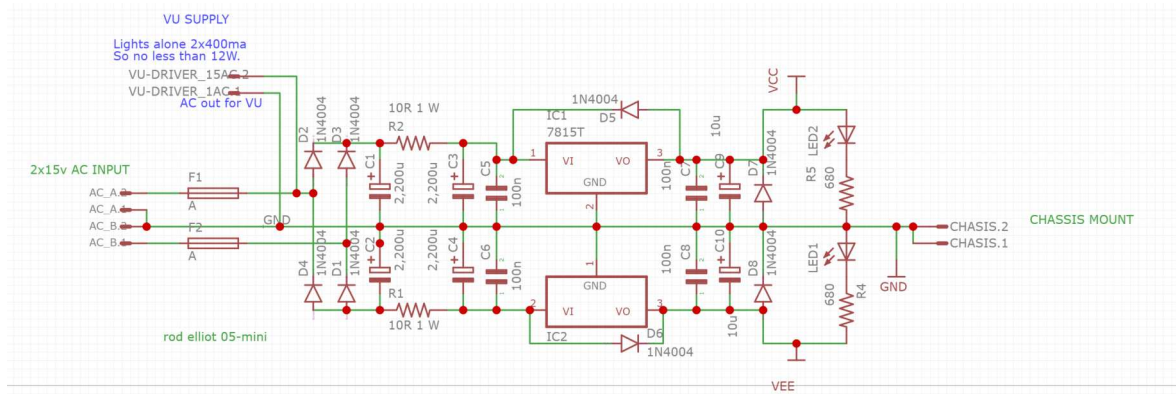


Figure 2.3: Power-supply schematic used by the preamplifier.

2.3.2 Transformer Selection and VA Estimate

The power transformer has a $15 - 0 - 15\text{ VAC}$ centre-tapped secondary and an approximate rating of 15 VA . Project 05 also identifies a $15 - 0 - 15\text{ VAC}$ transformer as a suitable source for a fixed $\pm 15\text{ V}$ preamplifier supply [1].

The approximate secondary-current rating corresponding to a 15 VA , 30 V end-to-end secondary is

$$I_{\text{sec}} = \frac{S}{V_{\text{sec}}} = \frac{15\text{ VA}}{30\text{ V}} \approx 0.50\text{ A}. \quad (2.1)$$

This calculation is a consistency estimate based on the nominal transformer rating; it is not a measured current value. A 0.50 A secondary rating provides substantial margin for a preamplifier whose active circuitry operates at comparatively low current.

2.3.3 Rectification

The centre tap of the transformer is connected to the circuit reference, and the two outer secondary connections feed the rectifier section. The rectifier uses discrete 1N4004 silicon diodes.

For a 15 VRMS half-secondary, the ideal sinusoidal peak is

$$V_{\text{pk}} = \sqrt{2}V_{\text{RMS}} \approx \sqrt{2} \times 15 \text{ V} \approx 21.2 \text{ V}. \quad (2.2)$$

Allowing approximately one silicon-diode forward drop during each conducting half-cycle gives a rough unloaded DC peak close to 20.5 V per rail. The actual voltage depends on mains voltage, transformer regulation, diode current, capacitor ripple, and circuit load; the value above is therefore a theoretical estimate only.

2.3.4 Smoothing Capacitors

Four 2200 μF electrolytic capacitors are installed in the power-supply section. These capacitors smooth the rectified positive and negative rails before linear regulation. Project 05 uses the same order of capacitance for this type of preamplifier supply [1].

For a capacitor-input rectifier, the approximate ripple relationship is

$$\Delta V \approx \frac{I_{\text{load}}}{f_{\text{ripple}}C}, \quad (2.3)$$

where I_{load} is the load current, f_{ripple} is the charging-ripple frequency, and C is the effective smoothing capacitance. Because the actual supply current was not measured, this relationship is included as a design equation rather than as a basis for a claimed measured ripple value.

2.3.5 Linear Regulation

After rectification and smoothing, the positive rail is regulated by a Texas Instruments LM7815 and the negative rail by a Texas Instruments LM7915. The design target is therefore

$$+15 \text{ V}, \quad 0 \text{ V}, \quad -15 \text{ V}.$$

The physical regulators are fitted with small heatsinks. The approximate regulator dissipation can be estimated from

$$P_{\text{reg}} = (V_{\text{in}} - V_{\text{out}}) I_{\text{load}}. \quad (2.4)$$

Because the load current and regulator input voltages were not formally measured, no numerical dissipation value is claimed for the completed unit.

2.3.6 Local Decoupling and Bypass Capacitors

Large smoothing capacitors alone are not sufficient to provide low supply impedance at high frequency. The final hardware therefore uses local 100 nF ceramic bypass capacitors together with 10 μ F electrolytic capacitors near the operational-amplifier supply points.

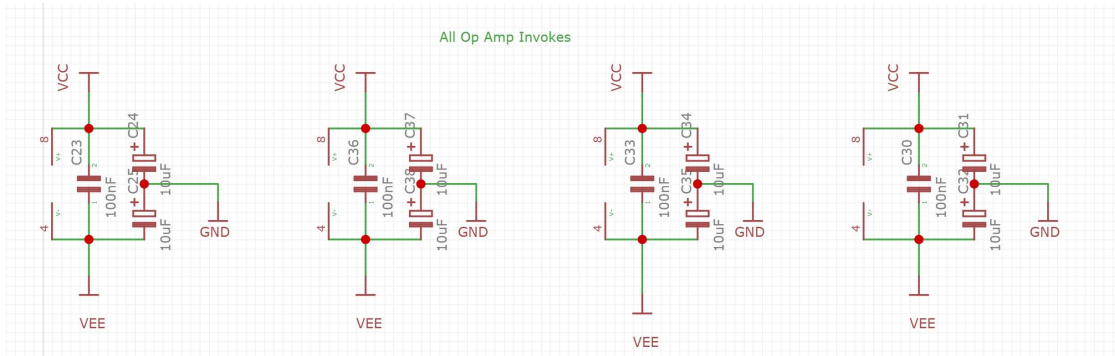


Figure 2.4: Local operational-amplifier supply decoupling and bypass arrangement.

The small ceramic capacitors provide a short high-frequency current path, while the larger electrolytic capacitors provide additional local energy storage at lower frequencies. This is particularly relevant to the LME49720, whose gain-bandwidth product is specified as 55 MHz [5].

2.4 Grounding and Safety Architecture

2.4.1 Signal Ground and Star Ground

Grounding is a critical part of a low-level analog audio system because the signal voltage is always defined relative to a return reference. Real PCB tracks and wires have finite resistance and inductance, so current flowing through a shared return path can create an unwanted voltage.

The principal relationship is

$$V_g = I_{\text{return}} Z_g, \quad (2.5)$$

where Z_g is the impedance of the shared ground path. If a power-supply or charging current shares the same path as a sensitive audio return, the resulting voltage can be added to the wanted signal.

The completed preamplifier uses a star-ground arrangement. The principal audio returns are brought to a dedicated chassis grounding point rather than relying on uncontrolled multiple chassis contacts.

2.4.2 Protective Earth and Chassis Bonding

The protective-earth conductor is connected directly to the same dedicated chassis grounding point used as the system star point. The metal enclosure is therefore bonded to protective earth.

This connection is distinct in purpose from the audio signal returns: protective earth is a safety connection to the conductive enclosure, while the signal-ground system establishes the electrical reference for the analog circuits. In the completed unit, the two systems meet at the controlled star point rather than at multiple arbitrary positions around the enclosure.

2.4.3 RCA Connector Isolation

The panel-mounted RCA connectors are electrically isolated from the metal chassis. Their outer conductors therefore do not establish additional direct chassis connections at the rear panel. Instead, the associated signal returns are routed through the intended grounding network to the common star point.

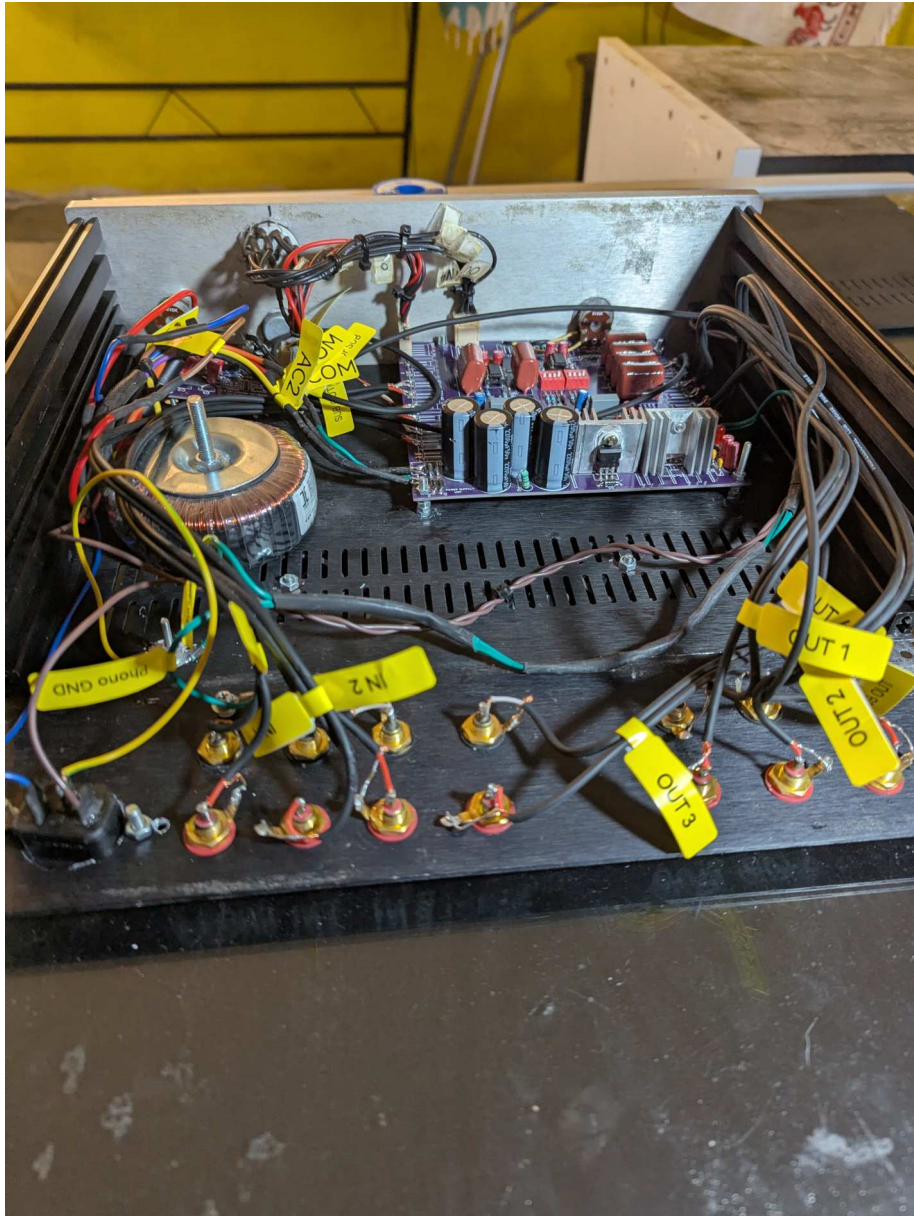


Figure 2.5: Internal rear-panel wiring showing the chassis-mounted RCA connections and labelled signal wiring.

This arrangement was important during troubleshooting because it reduced the number of uncontrolled conductive paths between signal ground and the metal enclosure.

2.5 Chapter Summary

The completed preamplifier uses a 15–0–15 VAC toroidal transformer, 1N4004 discrete rectifier diodes, four 2200 μF smoothing capacitors, LM7815/LM7915 regulators, and local 10 μF +100 nF decoupling. The nominal analog supply is $\pm 15\text{ V}$, although the rail voltages were not formally measured.

The metal enclosure is connected to protective earth at a dedicated chassis star point.

The signal returns are referenced to the same controlled point, while the RCA connectors are electrically isolated from the chassis. These decisions form the basis of the power and grounding architecture used by the remaining audio stages.

Chapter 3

Input Architecture and Phono Preamplifier

3.1 Input Architecture

The preamplifier provides three active source positions: one phono input and two conventional line-level inputs. The mechanical rotary selector has four positions, but only three are used in the completed unit; the remaining position is spare.

Each active source position is routed through the selector before entering the appropriate processing path. The two line inputs require no RIAA equalisation and therefore proceed directly to the line preamplifier. The phono input first passes through the dedicated Project 06 stage.

3.2 Input Selection

A front-panel rotary switch performs source selection. Both channels of the selected stereo source are switched together so that one left/right source pair is connected to the signal path at a time.

The implemented selector positions are:

Table 3.1: Implemented input-selector positions.

Position	Function
1	Phono input
2	Line input 1
3	Line input 2
4	Unused / spare

The physical routing between the rear-panel connectors and the selector was later identified as an important contributor to crosstalk. The final wiring therefore uses improved cable routing and coaxial signal cable on susceptible connections, as discussed in Chapter 6.

3.3 Line-Level Inputs

The two line-level inputs are intended for conventional analog sources such as digital-to-analog converters, media players, tuners, or other equipment with unbalanced line outputs.

The line inputs do not require frequency-dependent equalisation. Their main design requirement is therefore to deliver the selected signal to the line preamplifier with minimal unnecessary loading, interference, or coupling to other sources.

Because the inputs are unbalanced RCA connections, the signal conductor and its corresponding return are both important parts of the signal path. The internal wiring was therefore treated as part of the analog design rather than as a purely mechanical connection.

3.4 Moving-Magnet Phono Preamplifier

3.4.1 Reference Design

The phono stage is based on Elliott Sound Products Project 06, “Hi-Fi Phono Preamp (RIAA Equalisation)” [2]. This is distinct from the Project 88 line preamplifier and provides the additional gain and frequency-dependent equalisation required for vinyl playback.

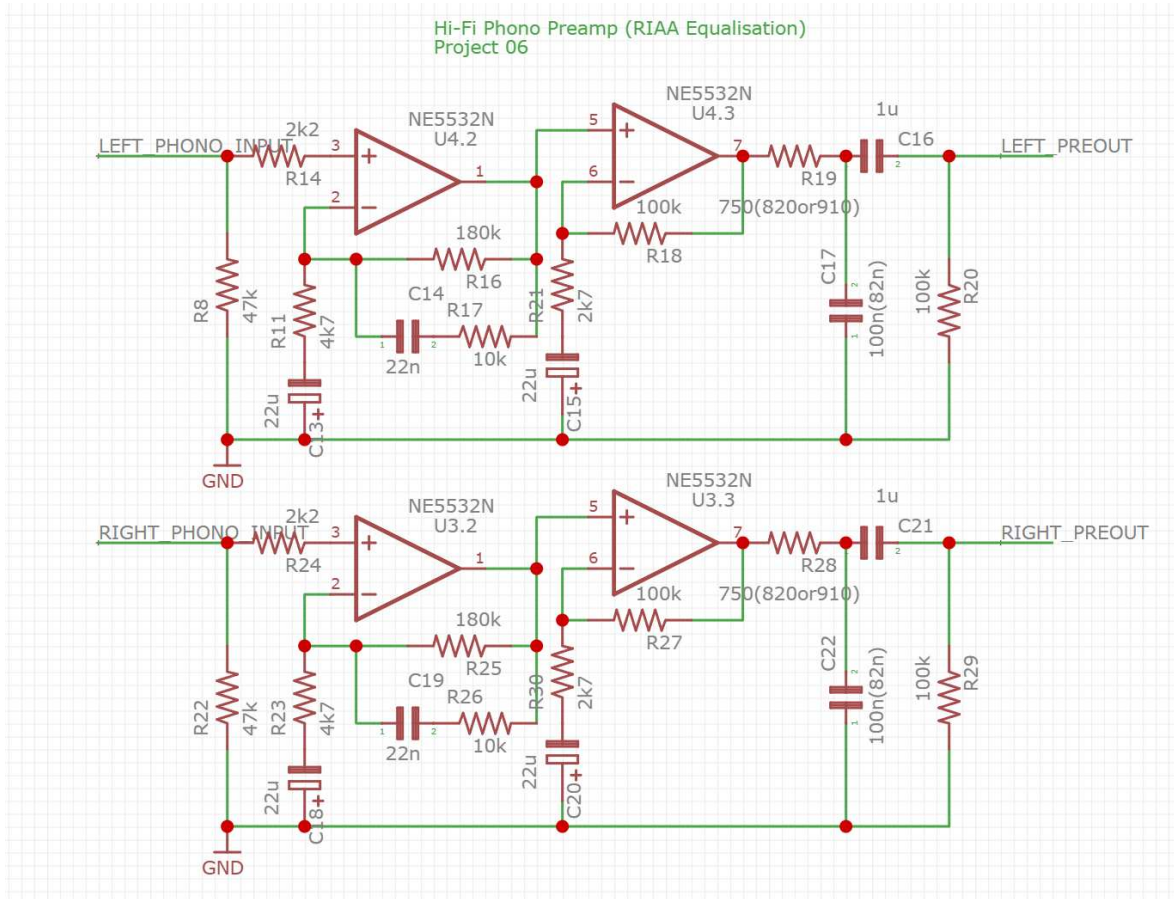


Figure 3.1: Implemented phono preamplifier and RIAA equalisation schematic.

3.4.2 Moving-Magnet Operation

Project 06 is intended for standard moving-magnet cartridges [2]. The completed preamplifier was used with a Rega Planar 1 turntable fitted with a moving-magnet cartridge.

A moving-magnet cartridge produces a signal that is considerably smaller than a normal line-level source. The phono stage must therefore provide substantial low-noise gain before the signal is passed to the line-level section.

Noise introduced at the phono input is especially important because it is amplified together with the wanted signal. For this reason, the phono path benefits from short wiring, suitable shielding, local decoupling, and physical separation from the transformer and mains conductors.

3.4.3 RIAA Equalisation

Vinyl records are produced using a frequency-dependent recording characteristic. During playback, the phono preamplifier applies the complementary RIAA response in order to

restore the intended spectral balance.

The principal RIAA playback time constants are

$$\tau_1 = 3180 \mu\text{s}, \quad \tau_2 = 318 \mu\text{s}, \quad \tau_3 = 75 \mu\text{s}. \quad (3.1)$$

Using

$$f = \frac{1}{2\pi\tau}, \quad (3.2)$$

the corresponding characteristic frequencies are approximately

$$f_1 \approx 50.05 \text{ Hz}, \quad f_2 \approx 500.5 \text{ Hz}, \quad f_3 \approx 2122 \text{ Hz}. \quad (3.3)$$

The Project 06 topology uses a combination of active and passive equalisation networks and is designed specifically for accurate RIAA playback [2]. The component values shown in the implemented schematic are therefore frequency-determining components and should be treated as part of the equalisation network rather than as arbitrary coupling values.

3.4.4 Operational-Amplifier Implementation

The original Project 06 documentation discusses several suitable low-noise audio operational amplifiers [2]. The final preamplifier uses LME49720 dual operational amplifiers in the phono section as part of the common op-amp strategy adopted for the completed unit.

The LME49720 is specified by Texas Instruments as a dual high-performance audio operational amplifier with a typical input-noise density of $2.7 \text{ nV}/\sqrt{\text{Hz}}$, a typical gain-bandwidth product of 55 MHz , and operation from supplies up to $\pm 17 \text{ V}$ [5]. The implemented $\pm 15 \text{ V}$ supply is therefore within its specified operating range.

3.5 Chapter Summary

The source-selection system provides three active inputs from a four-position rotary selector: Phono, Line 1, and Line 2. The two line-level sources are routed directly to the main preamplifier, while the phono signal first passes through a Project 06 moving-magnet RIAA stage.

The phono input was used with a Rega Planar 1 moving-magnet turntable source. Because the phono signal is the smallest signal handled by the unit, its wiring, grounding,

shielding, and supply decoupling are particularly important to the practical performance of the system.

Chapter 4

Line Preamplifier, Volume Control and Outputs

4.1 Project 88 Line Preamplifier

The main line-level signal-processing section is based on Elliott Sound Products Project 88, “High Quality Audio Preamp” [3]. The reference design provides multiple gain options and a control arrangement that can include both balance and volume adjustment.

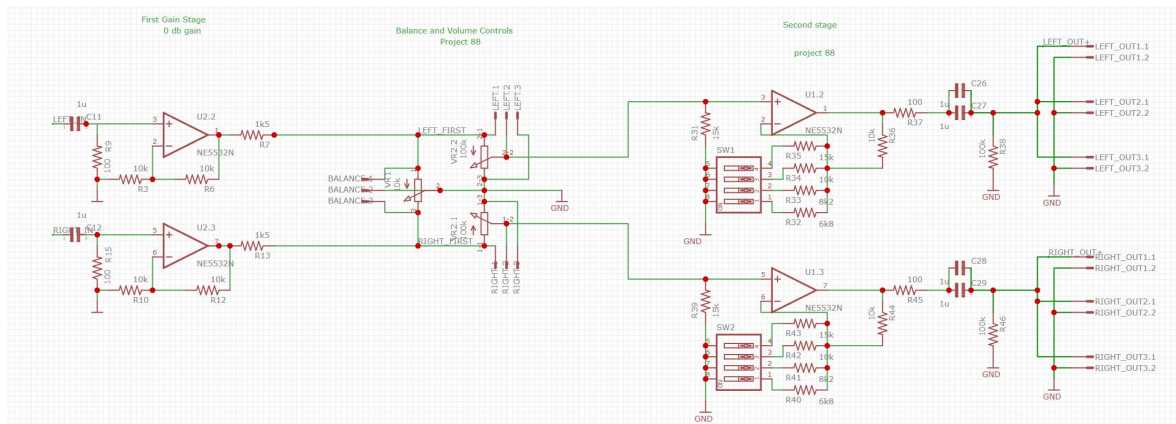


Figure 4.1: Line-level gain and signal-control schematic used in the project.

The completed preamplifier does not implement every optional feature of the reference design. In particular, only the 15 k Ω second-stage gain option is used and the balance control is bypassed.

4.2 Implemented Gain Configuration

Project 88 provides alternative second-stage gain-setting resistor values, including 15 k Ω , 10 k Ω , 8.2 k Ω , and 6.8 k Ω . The completed unit uses only the 15 k Ω option because it provides sufficient signal level for the associated power amplifier.

According to the published Project 88 gain-setting table, the 15 k Ω option corresponds to approximately 6.02 dB of gain in the second stage of the reference circuit [3].

The higher-gain resistor options were therefore not required in normal use. Retaining the lower available gain also preserves additional internal headroom for high-level line sources.

The total gain table published for the original Project 88 includes the loss of the reference balance-control network [3]. Because the balance control is bypassed in this implementation, that total-gain table is not copied directly as a measured or exact total-system specification here.

4.3 Volume Control

The completed preamplifier uses a 100 k Ω logarithmic stereo potentiometer as the main volume control. Both channels are adjusted by the same mechanical control.

The potentiometer acts as a variable attenuator. If the position-dependent attenuation is represented by k , the relationship can be written as

$$V_{\text{out}} = kV_{\text{in}}, \quad 0 \leq k \leq 1. \quad (4.1)$$

A logarithmic law is appropriate for an audio volume control because it provides a more useful relationship between control rotation and perceived level than a simple linear resistance law.

4.4 Balance-Control Bypass

Although Project 88 includes a balance-control arrangement, the completed preamplifier does not use a balance potentiometer. The balance section is bypassed and the unit therefore provides no front-panel left/right balance adjustment.

This is a deliberate implementation choice rather than a missing control. Project 88 explicitly allows the balance potentiometer to be omitted from the circuit [3]. The signal path in the completed hardware is therefore simplified compared with the full reference-control arrangement.

4.5 Main Line Outputs

The completed preamplifier provides three stereo line-output pairs. These outputs are connected in parallel to the same left- and right-channel output signals rather than being driven by three independent buffer stages.

When several external devices are connected simultaneously, their input impedances therefore appear in parallel. For three connected loads, the effective impedance is

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}. \quad (4.2)$$

This arrangement is suitable provided that the combined external load remains comfortably above the minimum load that the output stage can drive. The LME49720 is specified for high-fidelity operation with relatively low loads, including $600\,\Omega$ test conditions in the manufacturer data [5].

The line outputs are not intended to drive passive loudspeakers directly. They are intended for power amplifiers, active loudspeakers, recording equipment, headphone amplifiers, or other devices with line-level inputs.

4.6 Subwoofer Low-Pass Output

4.6.1 Project 155 Reference Circuit

The subwoofer-filter board is based on the low-pass section of Elliott Sound Products Project 155, “Variable High-Pass And Low-Pass Filters” [4]. The Project 155 low-pass circuit is a second-order active filter whose cut-off frequency is adjusted using a dual-gang control.

The filter board contains two independent mono channels. In the completed preamplifier, only one mono channel is used for the subwoofer output. The second identical mono channel is not connected.

4.6.2 Frequency-Setting Capacitors

The Project 155 low-pass filter uses three equal frequency-setting capacitors. The published reference values are $C_1 = C_2 = C_3 = 10\,\text{nF}$ [4]. In the completed subwoofer board, all three capacitors in the active channel were changed to

$$C_1 = C_2 = C_3 = 0.5\,\mu\text{F} = 500\,\text{nF}. \quad (4.3)$$

The implemented capacitance is therefore

$$\frac{500 \text{ nF}}{10 \text{ nF}} = 50 \quad (4.4)$$

times the original value. For unchanged resistor values, the characteristic frequency scales inversely with capacitance. Project 155 gives the original -3 dB frequencies as approximately 1.0 kHz at maximum potentiometer resistance, 1.9 kHz at the midpoint, and 11.3 kHz at minimum resistance [4].

Dividing these frequencies by 50 gives the theoretical range used in the completed subwoofer filter:

Table 4.1: Theoretical Project 155 low-pass frequencies after replacing all three 10 nF capacitors with 0.5 μ F.

Control condition	Reference value	Scaled value
Maximum resistance	1.0 kHz	20 Hz
Approx. midpoint	1.9 kHz	38 Hz
Minimum resistance	11.3 kHz	226 Hz

The theoretical adjustable range is therefore approximately 20 Hz to 226 Hz. These values are calculated from the published Project 155 scaling relationship and the installed capacitor values; they were not measured on the completed hardware.

4.6.3 Subwoofer Controls

The front panel provides a dedicated subwoofer cut-off-frequency control and a separate subwoofer level control. The filter frequency can therefore be adjusted independently of the main stereo signal, while the subwoofer-output amplitude can be matched to the associated low-frequency amplification system.

The subwoofer output remains a line-level signal and is intended to feed an active subwoofer or an external subwoofer power amplifier rather than a passive loudspeaker.

4.7 LME49720 Operational-Amplifier Selection

The final constructed unit uses LME49720 dual operational amplifiers. The LME49720 is specified by Texas Instruments for high-fidelity audio applications and provides low input-voltage noise, low specified distortion, a typical 55 MHz gain-bandwidth product, and operation from supplies up to ± 17 V [5].

The device is used from the nominal $\pm 15\text{ V}$ supply described in Chapter 2. In practical use, the installed devices operated normally in the completed preamplifier and no audible instability or abnormal behaviour was encountered.

The relatively high bandwidth of the LME49720 makes local supply bypassing and compact feedback paths particularly important. The 100 nF ceramic and $10\text{ }\mu\text{F}$ electrolytic local decoupling used in the project was therefore retained as part of the final implementation.

4.8 Chapter Summary

The main line preamplifier is based on Project 88 and uses the $15\text{ k}\Omega$ second-stage gain option only. A $100\text{ k}\Omega$ logarithmic stereo potentiometer provides the master volume control, while the reference balance-control function is bypassed.

Three stereo line outputs are connected in parallel. A separate Project 155 board provides an adjustable mono low-pass output for the subwoofer system. Using three $0.5\text{ }\mu\text{F}$ frequency-setting capacitors gives a theoretical cut-off range of approximately 20 Hz to 226 Hz . The final active devices are LME49720 dual audio operational amplifiers.

Chapter 5

PCB Design and Hardware Implementation

5.1 EDA Workflow

The schematic-development and PCB-layout workflow used two EDA packages. The circuit schematic was developed in Autodesk Eagle. The schematic was then exported or transferred into EasyEDA, where the final PCB layout was completed.

This workflow was chosen because the required PCB-layout capability in Autodesk Eagle was not available under the desired free-use arrangement, whereas EasyEDA provided the necessary PCB-layout tools without requiring a paid Eagle PCB licence for this project.

The final manufacturing files were generated from EasyEDA and the PCB was professionally fabricated by JLCPCB.

5.2 PCB Construction

The main board is a two-layer FR-4 PCB with a nominal thickness of 1.6 mm and 1 oz copper. The PCB predominantly uses through-hole components, including DIP operational amplifiers, film and electrolytic capacitors, resistors, connectors, regulators, and gain-setting components.

The two-layer construction provides routing flexibility while remaining straightforward to assemble manually. Through-hole construction also makes component replacement and troubleshooting convenient.

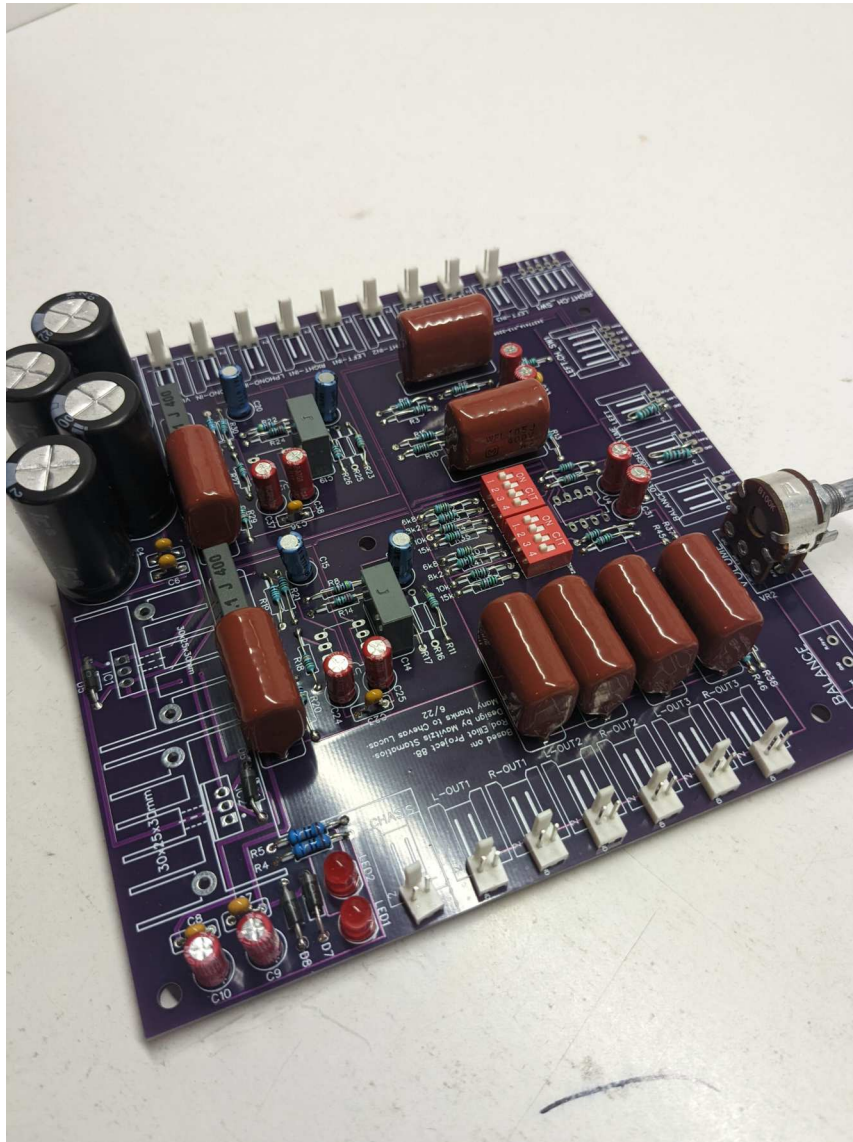


Figure 5.1: Assembled main preamplifier PCB.

5.3 Component Placement Strategy

Component placement was organised according to electrical function. Parts forming each operational-amplifier feedback network were kept close to the corresponding IC, while the main power-supply components were grouped within their own region of the board.

The large smoothing capacitors, rectifier components, regulators, and heatsinks occupy the power-supply area. The phono and line-level audio stages are located away from the transformer connection and the principal rectification paths as far as the board geometry allows.

The left- and right-channel circuitry follows similar physical arrangements so that the

stereo channels are implemented consistently.

5.4 Audio Signal Routing

Low-level audio routing was kept as direct as practical. Particular care was required around the phono input because interference coupled into this path is subsequently amplified by the phono stage.

The external wiring between the RCA connectors, selector, and PCB forms an extension of the PCB signal path. This became especially important during troubleshooting, when coupling between long adjacent input wires was found to contribute to audible crosstalk.

The final arrangement therefore uses coaxial cable on susceptible signal connections and avoids unnecessarily long parallel runs between unrelated input signals.

5.5 Power-Supply Routing

The rectifier and smoothing-capacitor paths carry charging currents that are larger and more impulsive than the small-signal currents in the audio sections. These paths were therefore treated separately from the most sensitive signal routing.

After regulation, the +15 V and −15 V rails are distributed to the active stages. Local bypass capacitors provide a short return path for rapid current changes near the operational amplifiers instead of forcing high-frequency current to travel through the complete power-distribution network.

5.6 Ground and Return-Path Design

The PCB and chassis grounding were considered as one system. Sensitive signal returns were prevented from relying on random chassis contacts, and the RCA connectors were insulated from the enclosure.

The principal circuit grounds are brought to the intended star-ground point on the chassis. This arrangement provides a defined relationship between signal ground, power-supply ground, chassis, and protective earth.

The ground system was refined during troubleshooting because the physical path taken by a return current can be as important as the schematic symbol used to represent that connection.

5.7 PCB Layer Layout and Ground Distribution

The two-layer PCB layout was designed so that the ground connections of the different functional sections are returned toward a common grounding region rather than being connected through uncontrolled series return paths. The top- and bottom-layer views are shown in Figure 5.2.

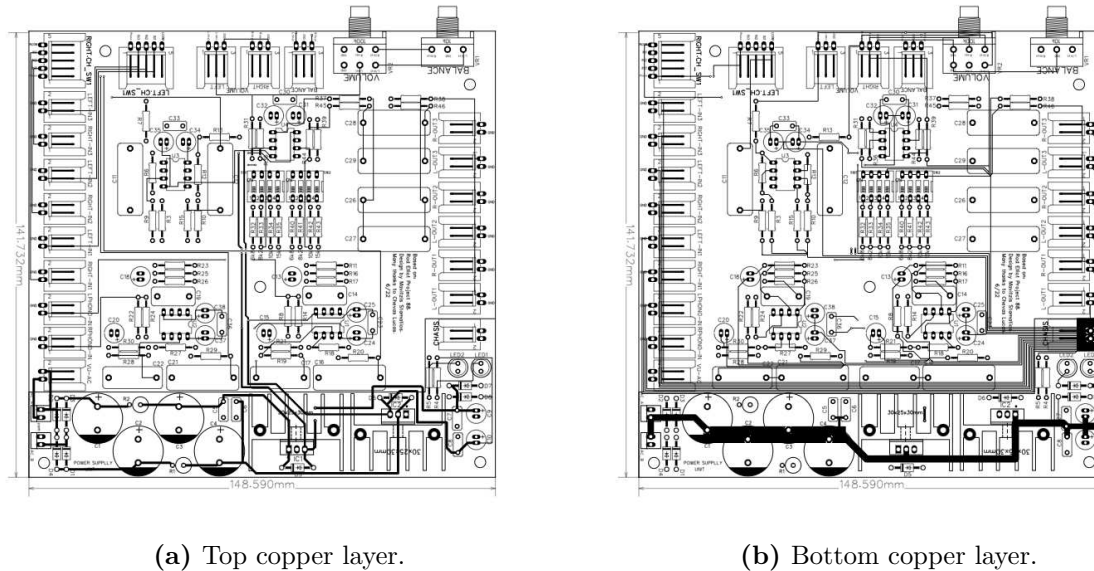


Figure 5.2: Top and bottom PCB copper layers showing the ground-routing arrangement of the main preamplifier board.

Inspection of both copper layers is useful because the complete return-current structure cannot be understood from either layer independently. Ground connections associated with the audio stages, power supply, input and output sections are routed through the PCB toward the common grounding connection of the board.

The PCB ground connection is subsequently connected to the dedicated chassis star-ground point described in Chapter 2. The chassis star point also provides the controlled connection to protective earth. In this way, the PCB routing and chassis wiring form a single grounding architecture rather than relying on multiple uncontrolled connections to the metal enclosure.

The complete layer artwork is reproduced at larger scale in Appendix C, where the routing and ground connections can be examined in greater detail.

5.8 Component Assembly

The fabricated boards were assembled manually. Smaller passive components were installed before many of the larger parts, followed by the larger film capacitors, electrolytic

capacitors, connectors, regulator heatsinks, and integrated-circuit sockets.

The LME49720 operational amplifiers were installed in sockets. This made the ICs easier to replace during development and avoided repeated soldering and desoldering of the PCB itself.

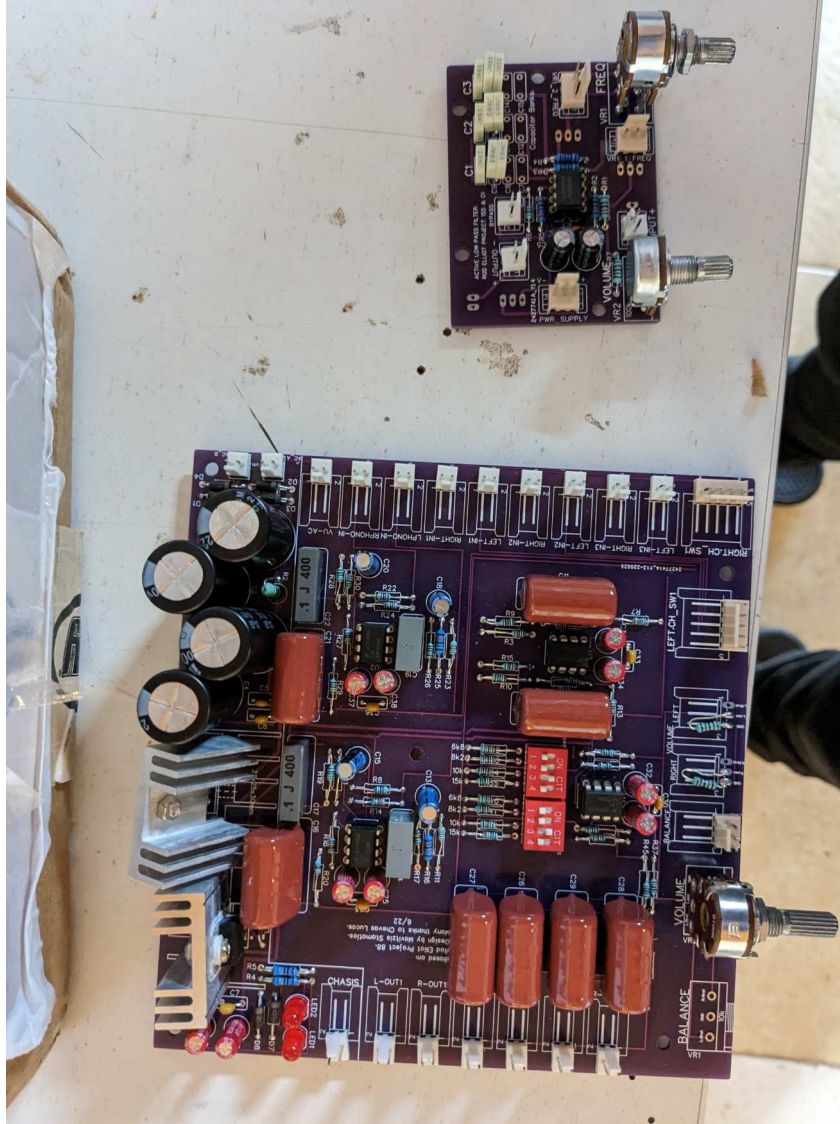


Figure 5.3: Main preamplifier PCB together with the separate low-pass-filter PCB.

The smaller PCB implements the variable low-pass-filter function based on Project 155. The board contains two mono filter channels, although only one is used in the completed subwoofer path.

5.9 Development and Construction Process

The hardware was developed iteratively. Early work included the physical planning of controls and circuitry before the final enclosure wiring was completed.

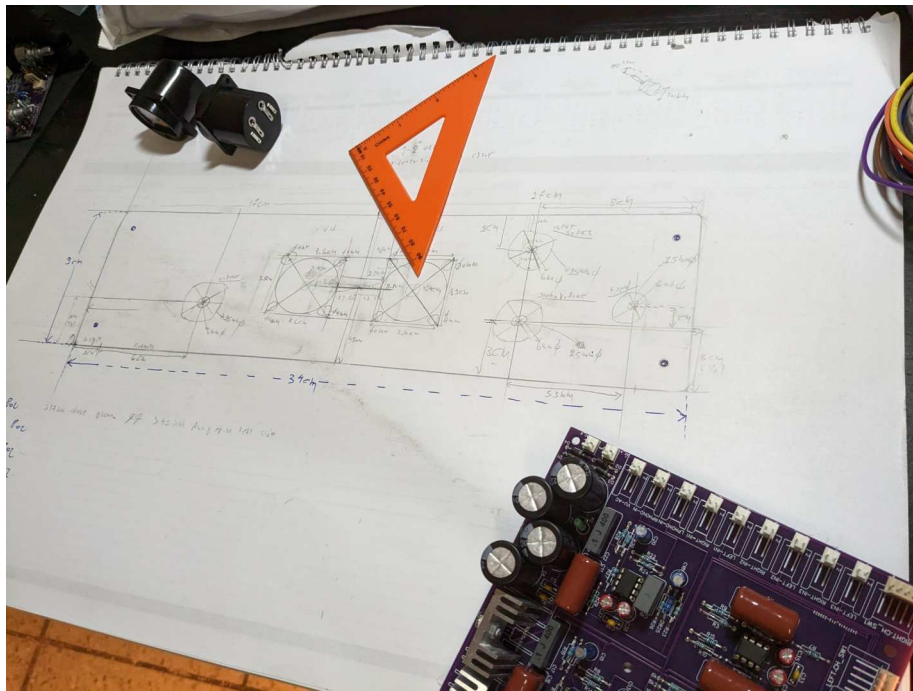


Figure 5.4: Early design and mechanical-planning stage with the main PCB and enclosure layout sketch.

The project then progressed through bench assembly and wiring before final installation in the enclosure.

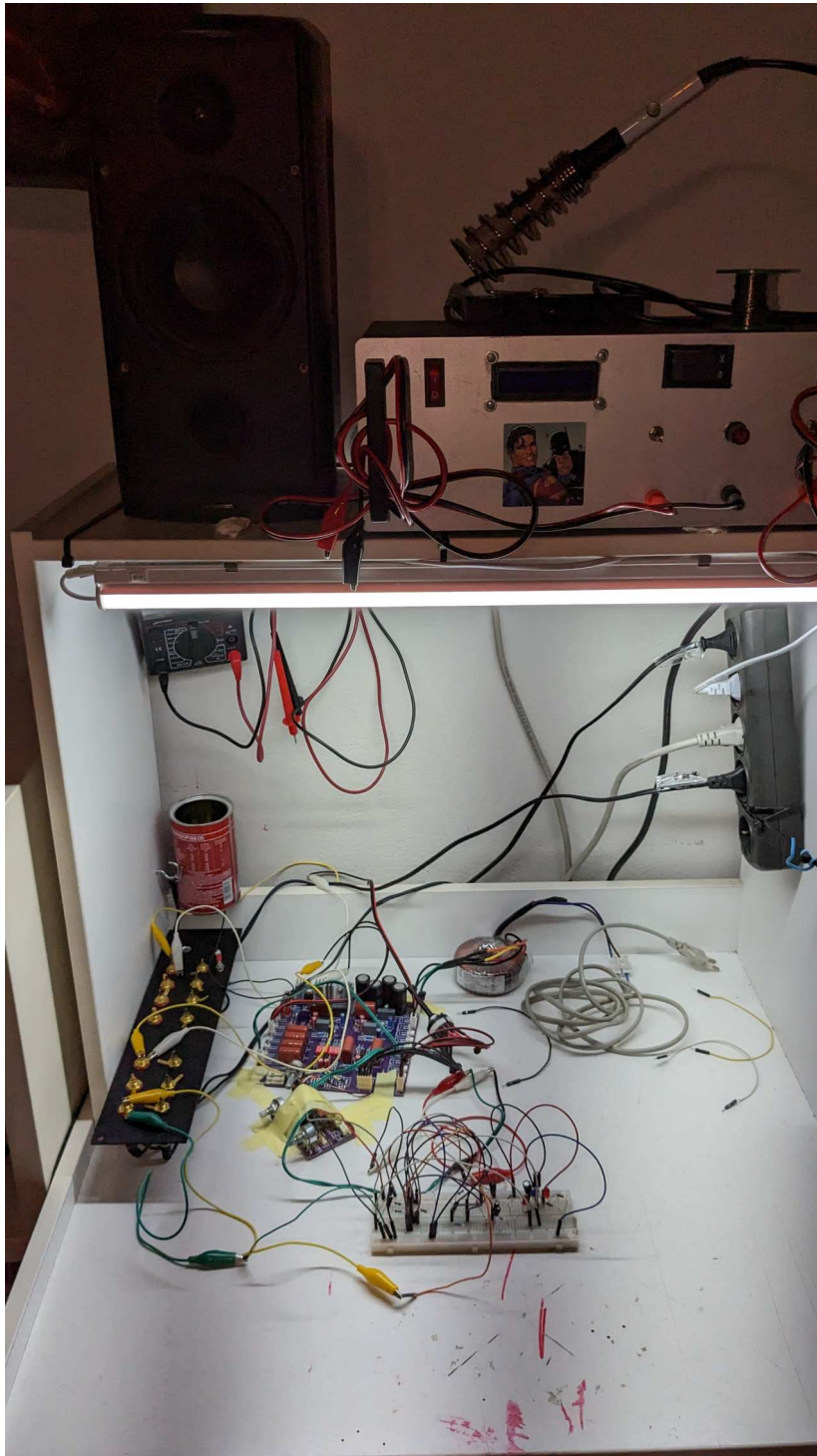


Figure 5.5: Bench wiring and integration stage during construction of the preamplifier.

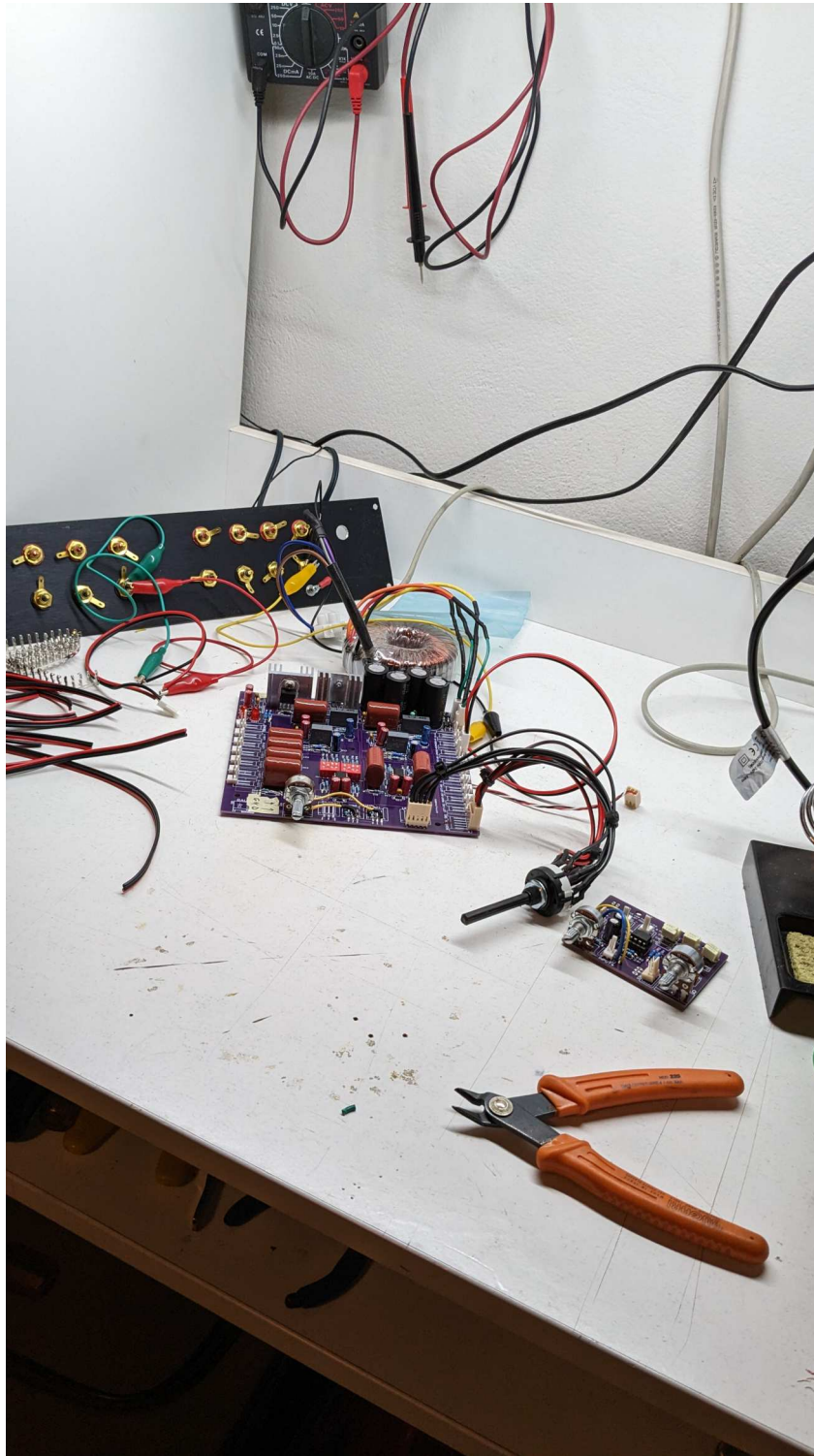


Figure 5.6: PCB and control interconnection work during the construction process.

5.10 Mechanical Integration

The completed electronics were installed in a ventilated metal enclosure. The toroidal transformer was mounted to the chassis base, while the main PCB and filter PCB were

mounted on spacers to prevent contact between soldered conductors and the metal enclosure.

The transformer and its associated AC wiring were kept away from low-level signal connections as far as practical. The front-panel potentiometers and selector were connected to the internal boards using labelled wiring, while the rear-panel RCA sockets were mounted mechanically to the chassis but insulated electrically from it.

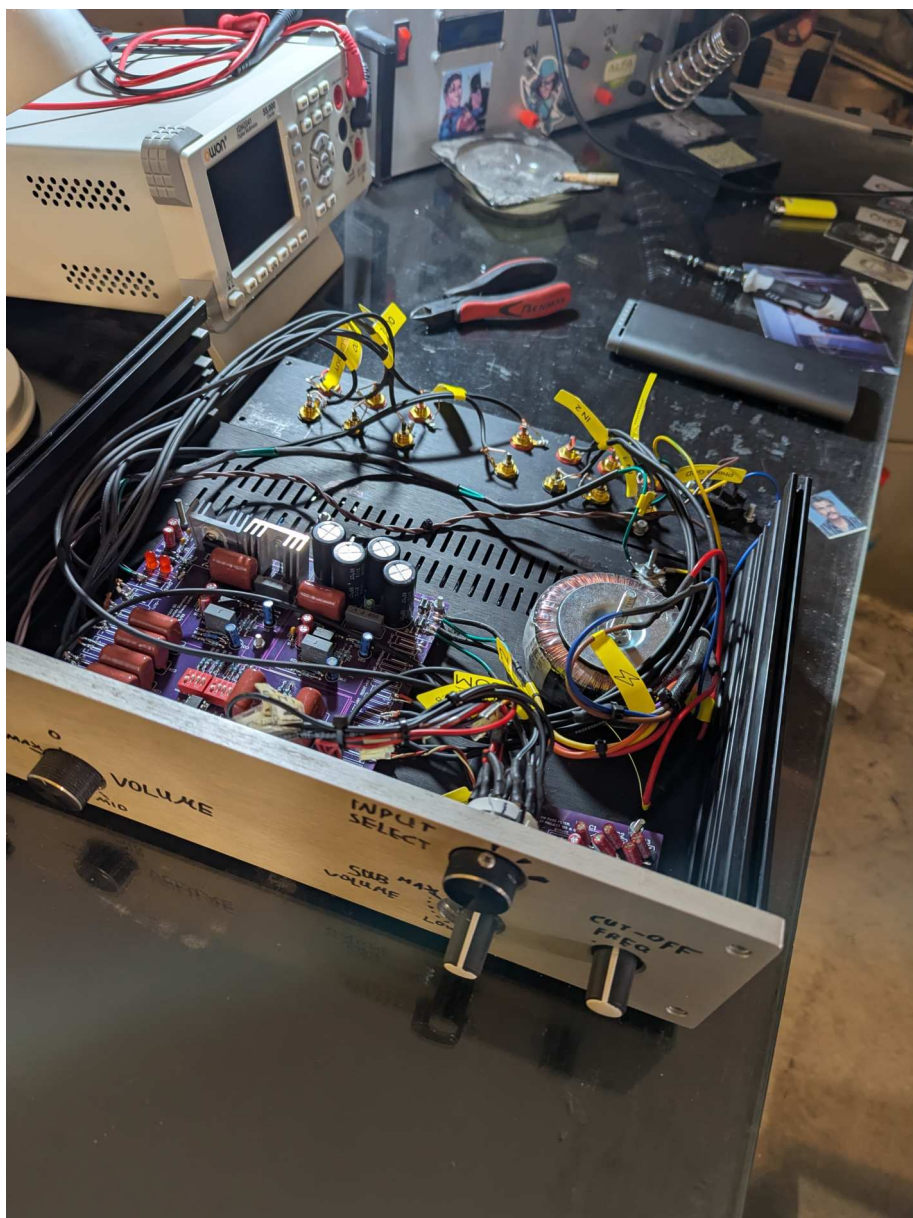


Figure 5.7: Completed preamplifier with the enclosure open during inspection.

The internal arrangement reflects a compromise between electrical separation, mechanical accessibility, enclosure dimensions, and the position of the front- and rear-panel controls.

5.11 Final Hardware Configuration

The final hardware combines the main preamplifier PCB, the Project 155 low-pass-filter PCB, the toroidal transformer, the front-panel controls, and the rear-panel input/output connectors in one self-contained unit.

The main functional characteristics of the completed hardware are summarised in Table 5.1.

Table 5.1: Final implemented configuration.

Function	Implemented configuration
Active inputs	Phono, Line 1, Line 2; fourth selector position unused
Phono stage	ESP Project 06, moving-magnet RIAA preamplifier
Line preamplifier	ESP Project 88
Line-stage gain option	15 k Ω option only
Master volume	100 k Ω logarithmic stereo potentiometer
Balance control	Not fitted; balance section bypassed
Main outputs	Three parallel stereo line-output pairs
Subwoofer filter	ESP Project 155 low-pass section, one mono channel used
Subwoofer capacitors	$C_1 = C_2 = C_3 = 0.5 \mu\text{F}$
Calculated subwoofer range	Approximately 20 Hz to 226 Hz
Operational amplifiers	LME49720 dual audio op-amps
Transformer	15 – 0 – 15 VAC, approximately 15 VA
Rectifier	Discrete 1N4004 diodes
Smoothing capacitors	Four 2200 μF electrolytics
Regulators	LM7815 and LM7915
Nominal analog rails	$\pm 15 \text{ V}$
Local decoupling	10 μF electrolytic + 100 nF ceramic
PCB construction	Two-layer, 1.6 mm FR-4, 1 oz copper
Grounding	Chassis star point shared by signal ground and protective-earth bond
RCA mounting	Mechanically chassis-mounted, electrically isolated

5.12 Chapter Summary

The schematic was developed in Autodesk Eagle and transferred to EasyEDA for PCB layout and production preparation. The main PCB was manufactured as a two-layer 1.6 mm FR-4 board with 1 oz copper and was assembled manually using predominantly through-hole components.

The completed unit integrates the main PCB, the separate subwoofer-filter board, the internal transformer, front-panel controls, rear-panel isolated RCA connectors, star-ground wiring, and protective-earth chassis bonding.

Chapter 6

Troubleshooting and Design Optimisation

6.1 Overview

Several practical issues became apparent only after the electronic circuits were connected to the panel-mounted controls and installed in the enclosure. The principal problems were audible coupling between input sources and mains-related hum.

The troubleshooting process was qualitative. The unwanted signals were identified by listening rather than by oscilloscope, FFT, or calibrated noise measurement. The discussion in this chapter therefore describes observed behaviour and implemented corrections without assigning unmeasured numerical performance values.

6.2 Input Crosstalk

6.2.1 Observed Behaviour

In the initial internal-wiring arrangement, a signal connected to one source input could be heard weakly when another input was selected. This indicated unwanted coupling between conductors that were intended to remain isolated by the source-selection system.

6.2.2 Likely Coupling Mechanism

The original installation contained several unbalanced signal wires routed near one another over a significant distance. Adjacent conductors form a small parasitic capacitance, allowing an AC signal on one conductor to couple into another even when no

direct galvanic connection exists.

The effect becomes more significant for long parallel runs and high-impedance receiving nodes. The problem was therefore associated with the complete interconnection system, not only with the PCB traces.

6.2.3 Implemented Correction

The susceptible input wiring was replaced with coaxial audio cable. The coaxial shield provides electrostatic shielding around the signal conductor, while the cable geometry also creates a controlled signal/return path.

Cable routing was additionally improved so that unrelated inputs were not kept together unnecessarily. After these changes, the audible bleed between unselected inputs was substantially reduced during normal operation.

6.3 Mains-Related Hum

6.3.1 Observed Behaviour

Audible hum was also encountered during development. The interference was identified by ear as mains-related hum rather than by a frequency-domain measurement. It is therefore described as mains-related or mains-frequency hum rather than as a measured 50 Hz component.

6.3.2 Possible Sources

Several mechanisms can introduce mains-related interference inside a self-contained preamplifier. These include magnetic coupling from the power transformer, electric-field coupling from mains conductors, coupling from the transformer secondary wiring, and ground-voltage differences caused by rectifier and smoothing-capacitor charging currents.

The phono stage is particularly sensitive because it operates with a much smaller input signal than the line-level stages.

6.3.3 Implemented Corrections

The internal wiring was rearranged to increase the separation between mains-related conductors and low-level audio connections. Sensitive signal paths were moved away

from the transformer where practical, and coaxial cable was used where additional shielding was beneficial.

The grounding arrangement was also revised so that the RCA connector shells did not establish unintended chassis contacts and the principal returns met at the intended chassis star point.

These changes reduced the audible hum in normal use and demonstrated that a regulated power supply alone cannot prevent interference if the physical wiring and grounding arrangement are poor.

6.4 Chassis and Grounding Optimisation

The rear-panel RCA connectors are mechanically mounted to the metal panel but are electrically insulated from it. This prevents each connector from creating its own direct signal-ground-to-chassis connection.

Instead, the signal returns are brought to the controlled grounding network, and the protective-earth conductor is bonded directly to the dedicated chassis star point. This arrangement made the grounding system easier to understand and reduced uncontrolled parallel return paths.

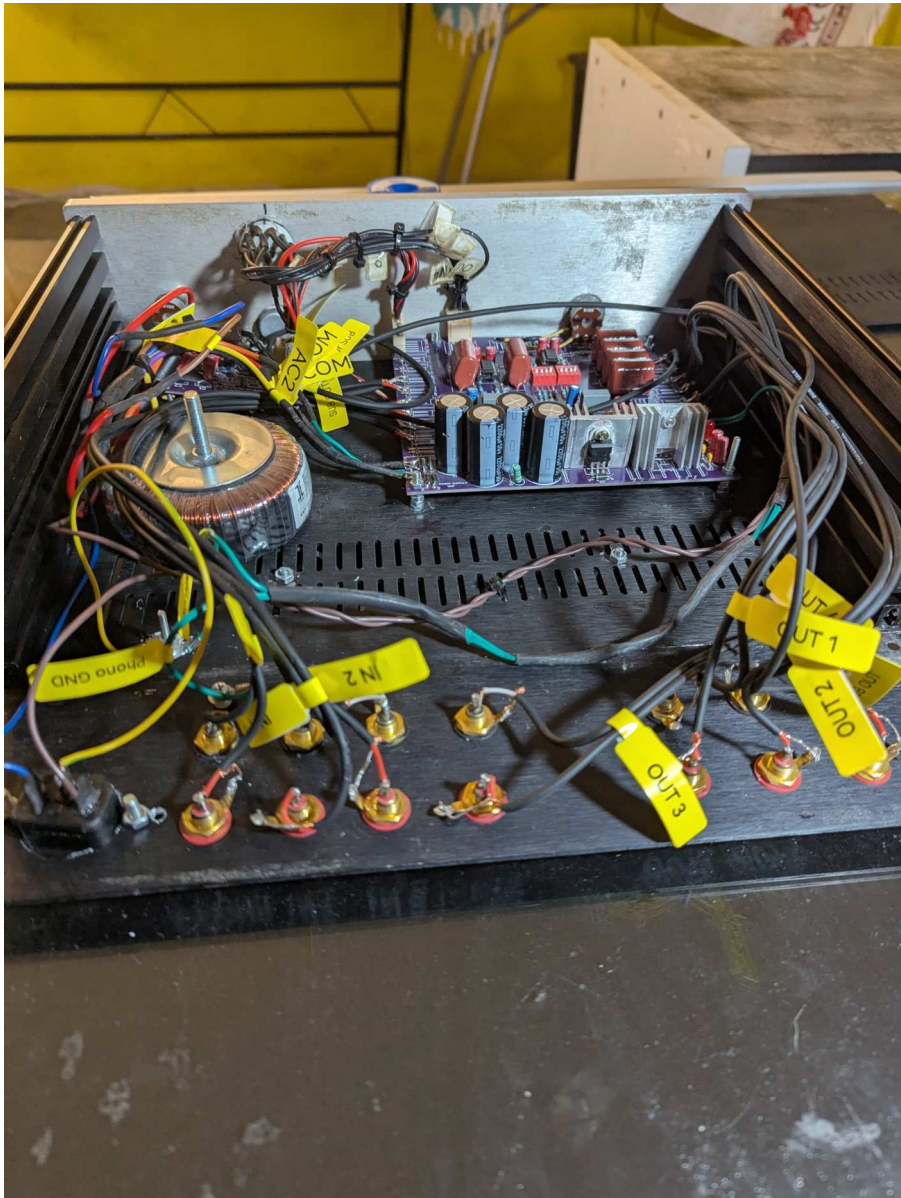


Figure 6.1: Rear-panel and internal wiring during inspection of grounding and signal routing.

6.5 Operational-Amplifier Stability and Decoupling

The final LME49720 devices have substantially wider bandwidth than older traditional audio operational amplifiers. High-bandwidth devices are more sensitive to parasitic inductance, long feedback paths, supply impedance, and capacitive loading.

No audible instability was encountered with the final implementation. Local 100 nF ceramic and 10 μ F electrolytic bypassing was nevertheless retained because it provides a low-impedance local supply path and follows good practice for a high-bandwidth audio operational amplifier.

6.6 Gain Choice and Source Compatibility

Different line sources can provide different nominal output voltages. The Project 88 circuit offers several gain-setting options for this reason [3].

In the completed system, the 15 k Ω second-stage option provided sufficient output for the associated power amplifier, so the higher-gain options were not required. This simplified normal operation and reduced the need for unnecessary internal gain.

6.7 Limitations of the Evaluation

The completed preamplifier was functionally tested and used for listening, but formal quantitative characterisation was not performed. In particular, no calibrated measurements were recorded for:

- total harmonic distortion or THD+N;
- signal-to-noise ratio;
- channel crosstalk;
- absolute frequency response;
- RIAA tracking error;
- exact line-stage gain;
- clipping level;
- output impedance;
- power-supply ripple; or
- exact positive and negative regulated rail voltages.

For this reason, the report does not present reference-design measurements as if they were measurements of the completed hardware. Published data from the reference designs and device datasheets are cited only to explain component selection and theoretical operation.

6.8 Chapter Summary

The most important practical improvements were made to the interconnection and grounding rather than to the basic circuit topology. Coaxial signal cable reduced audible input crosstalk, while improved physical separation and a controlled star-ground arrangement reduced mains-related hum.

The completed LME49720 implementation operated normally with the selected local decoupling. The 15 k Ω Project 88 gain option was retained because it provided sufficient level for the intended power amplifier.

Chapter 7

Final Discussion, Conclusions and Future Improvements

7.1 Achievement of Project Objectives

The completed hardware achieved the principal functional objectives defined for the project. It provides two line-level inputs, one moving-magnet phono input, rotary source selection, a master stereo volume control, three stereo line outputs, and an adjustable low-pass subwoofer output.

The unit is fully self-contained, with its own transformer, rectification, smoothing, $\pm 15\text{ V}$ regulation, active audio circuitry, controls, and input/output connectors in one metal enclosure.

The phono section allows direct connection of a moving-magnet turntable such as the Rega Planar 1 used with the project. The Project 88 line stage provides sufficient signal level using the $15\text{ k}\Omega$ gain option, and the main output signal can be distributed to as many as three line-level devices through the parallel output connectors.

7.2 Successful Design Decisions

Several design decisions were particularly useful in the completed system. The regulated symmetrical supply provides a simple common architecture for the analog stages. Socketed LME49720 devices allow service or replacement without modifying the PCB, while local decoupling supports stable operation of the high-bandwidth op-amps.

The use of a separate low-pass-filter PCB keeps the optional subwoofer function modular. The electrically isolated RCA connectors and controlled star-ground connection provide a clear grounding architecture, while the metal enclosure provides mechanical protection

and electromagnetic shielding.

The use of coaxial internal signal cable after troubleshooting was another important practical improvement because it addressed coupling that was not obvious from the schematic alone.

7.3 Limitations of the Present Design

The principal limitation of the present hardware is the amount of point-to-point wiring between the PCBs, front-panel controls, and rear-panel connectors. These wires increase assembly effort and create opportunities for capacitive and electromagnetic coupling.

The mains transformer is also located inside the same enclosure as the phono and line-level circuitry. Although the final routing and shielding produced acceptable practical operation, greater physical separation would reduce the potential for transformer-field coupling.

The three main line outputs are connected in parallel rather than being individually buffered. This is simple and effective for high-impedance loads, but the total connected load decreases when several low-impedance devices are connected simultaneously.

Finally, the present report cannot provide measured distortion, noise, frequency-response, or RIAA-tracking results because such measurements were not recorded when the unit was completed.

7.4 Recommended Future Improvements

A future revision could move the source-selection circuitry closer to the rear-panel input connectors. Relay-based input switching would allow the front-panel selector to carry only control signals while the sensitive audio paths remain short.

The transformer could be separated from the low-level audio circuitry by a metal partition or placed in an external power-supply enclosure. Either approach would increase magnetic and electrical separation from the phono stage.

A redesigned PCB could provide dedicated local bypass-capacitor footprints immediately adjacent to every operational-amplifier supply pair, more structured connectors for panel wiring, and clearer physical separation between the rectifier/smoothing region and the phono section.

The three main outputs could also be individually buffered if completely independent loading were required. The currently unused second Project 155 mono filter channel could be retained as a spare or repurposed in a future revision.

A future measurement campaign could characterise the final hardware using an audio signal generator and oscilloscope or audio analyser. Useful measurements would include line-stage gain, clipping level, frequency response, RIAA tracking, noise, crosstalk, power-supply ripple, and output impedance.

7.5 Conclusions

This project resulted in the successful construction of a complete standalone analog audio preamplifier based on established reference circuits but adapted to a specific practical implementation.

The system combines a Project 06 moving-magnet RIAA stage, a Project 88 line preamplifier, a Project 05-style regulated symmetrical power supply, and a Project 155 variable low-pass filter. The final active devices are LME49720 dual operational amplifiers operating from a nominal $\pm 15\text{ V}$ supply.

The project demonstrated that the physical implementation of an analog audio system is as important as the schematic itself. Input crosstalk and mains-related hum were improved mainly through changes to internal wiring, coaxial shielding, connector isolation, cable routing, and the grounding architecture.

The final preamplifier therefore represents a complete engineering exercise in analog audio electronics, including circuit selection, power-supply design, schematic capture, PCB design, component assembly, enclosure integration, grounding, troubleshooting, and final system implementation.

Complete Circuit Schematics

This appendix reproduces the principal schematics used in the completed preamplifier. The same figures are included in the relevant chapters so that they can be read alongside the technical discussion; they are repeated here for convenient complete reference.

A.1 Power-Supply Schematic

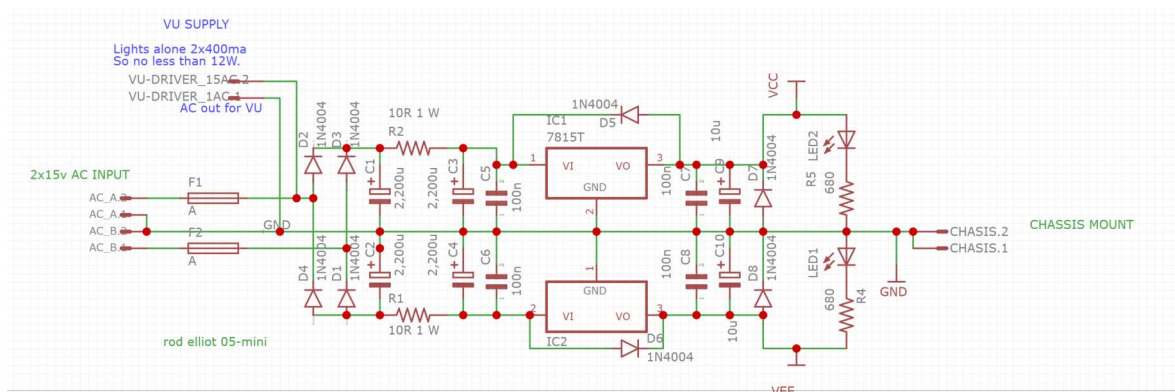


Figure A.1: Complete power-supply schematic.

A.2 Phono Preamplifier Schematic

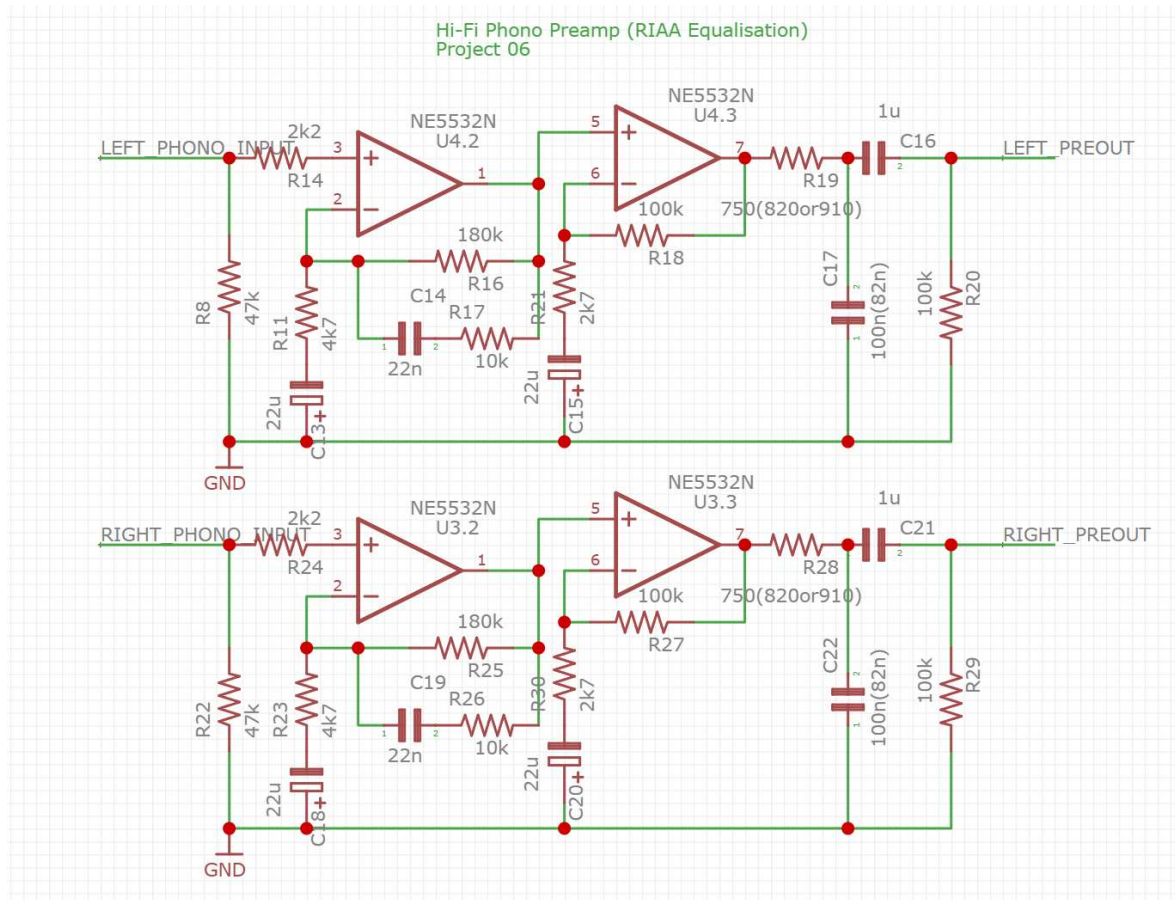


Figure A.2: Complete moving-magnet phono preamplifier and RIAA equalisation schematic.

A.3 Line Preamplifier and Gain Schematic

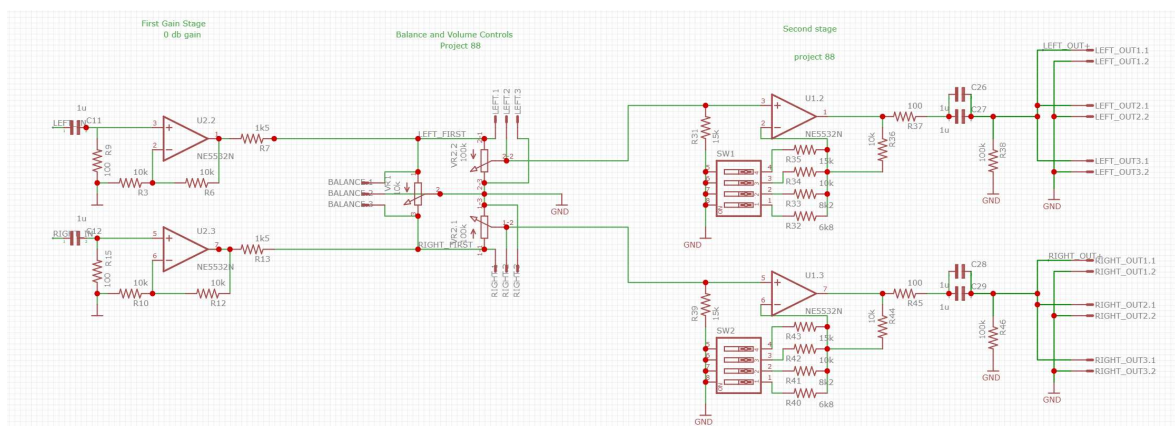


Figure A.3: Complete line-level gain and signal-processing schematic.

A.4 Operational-Amplifier Decoupling Schematic

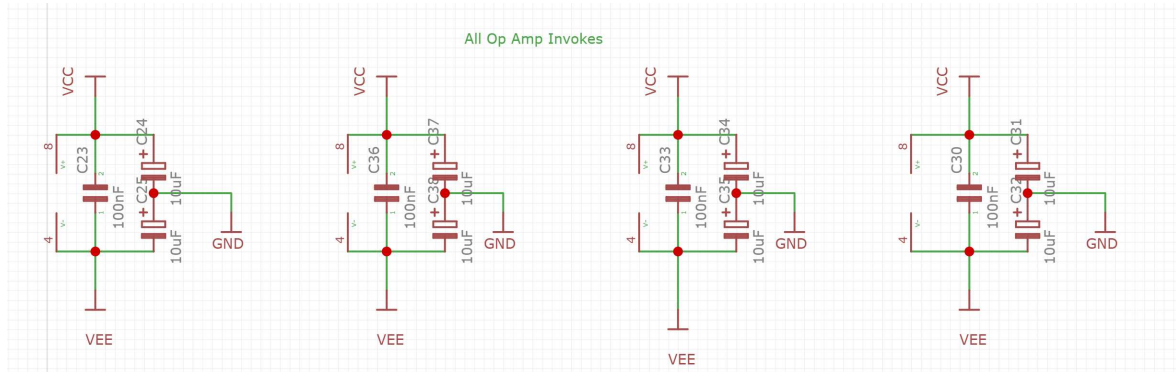


Figure A.4: Operational-amplifier local decoupling and bypass arrangement.

Appendix B

Project Photographs and Hardware Documentation

This appendix collects the principal construction photographs in one place. Several of these figures are also used in the main chapters where they support a specific technical discussion.

B.1 Completed Preamplifier



Figure B.1: Front view of the completed standalone preamplifier.

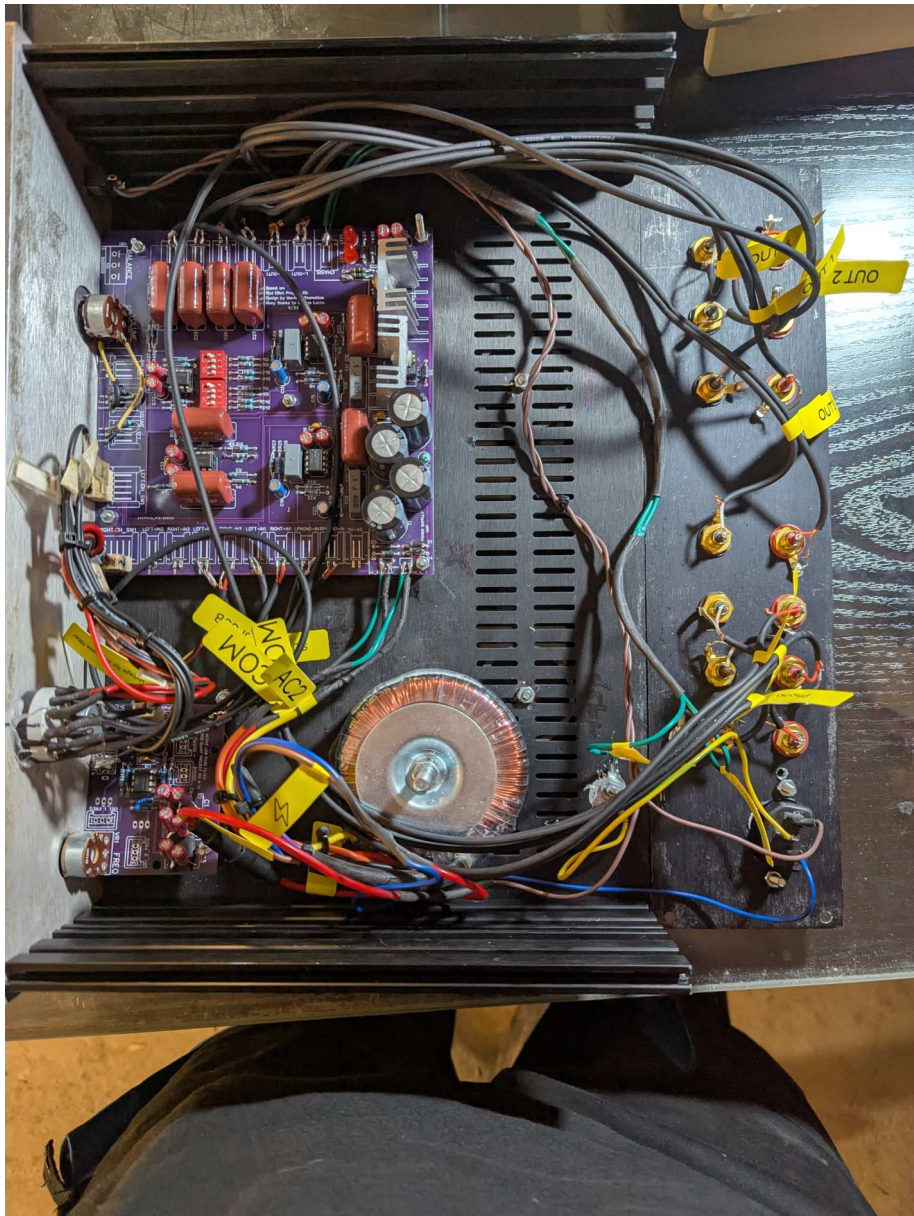


Figure B.2: Top-down internal view of the completed preamplifier.

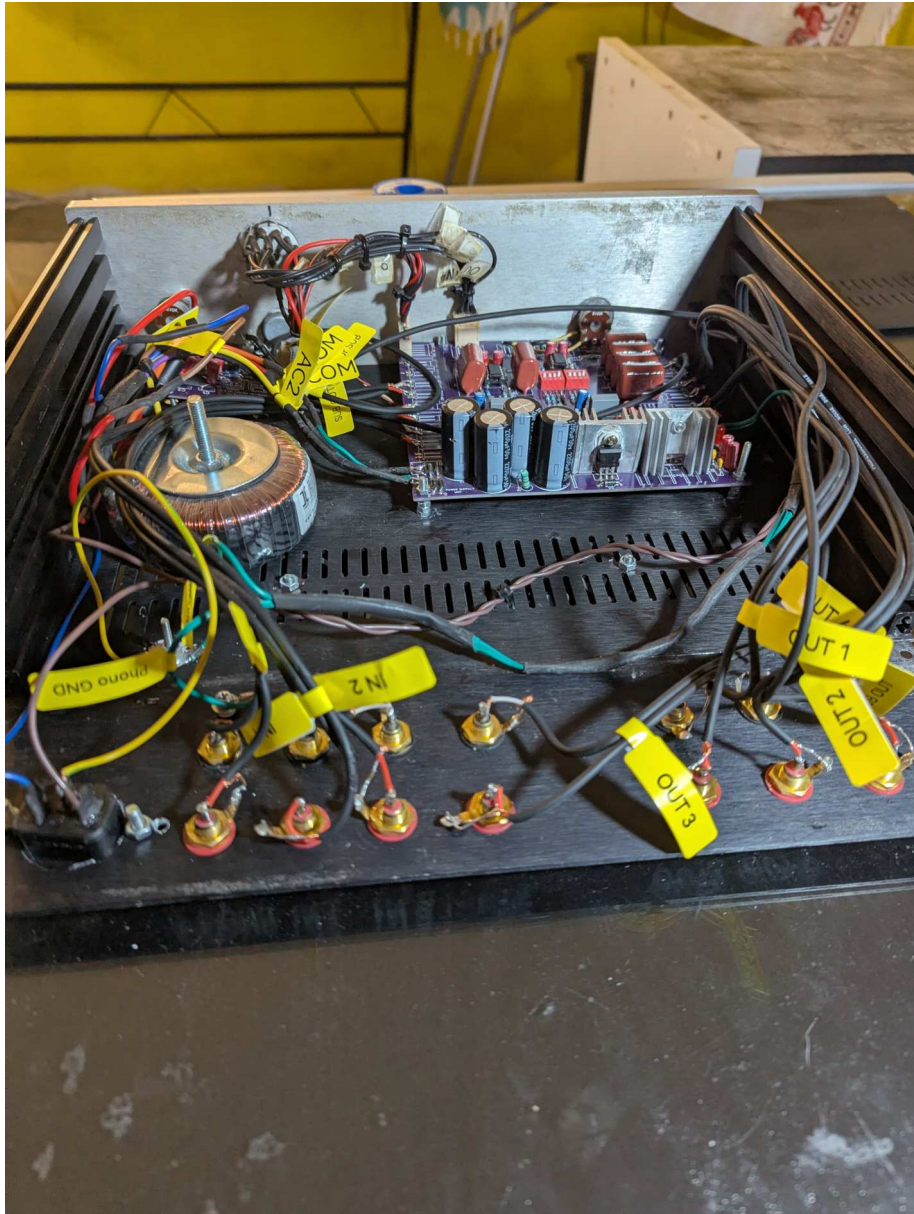


Figure B.3: Rear-panel connectors, labelled wiring, transformer, and main PCB inside the enclosure.

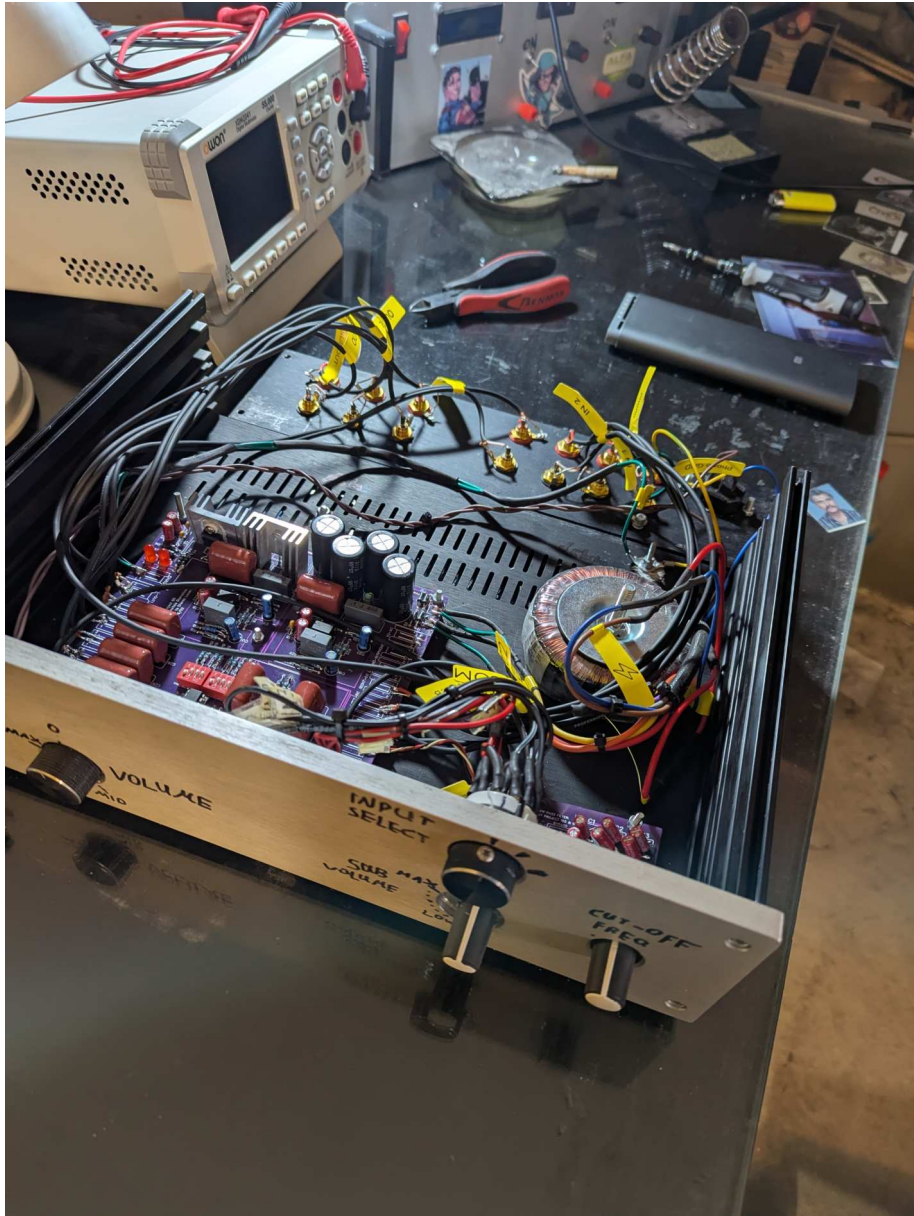


Figure B.4: Oblique view of the completed preamplifier with the enclosure open.

B.2 Printed Circuit Boards

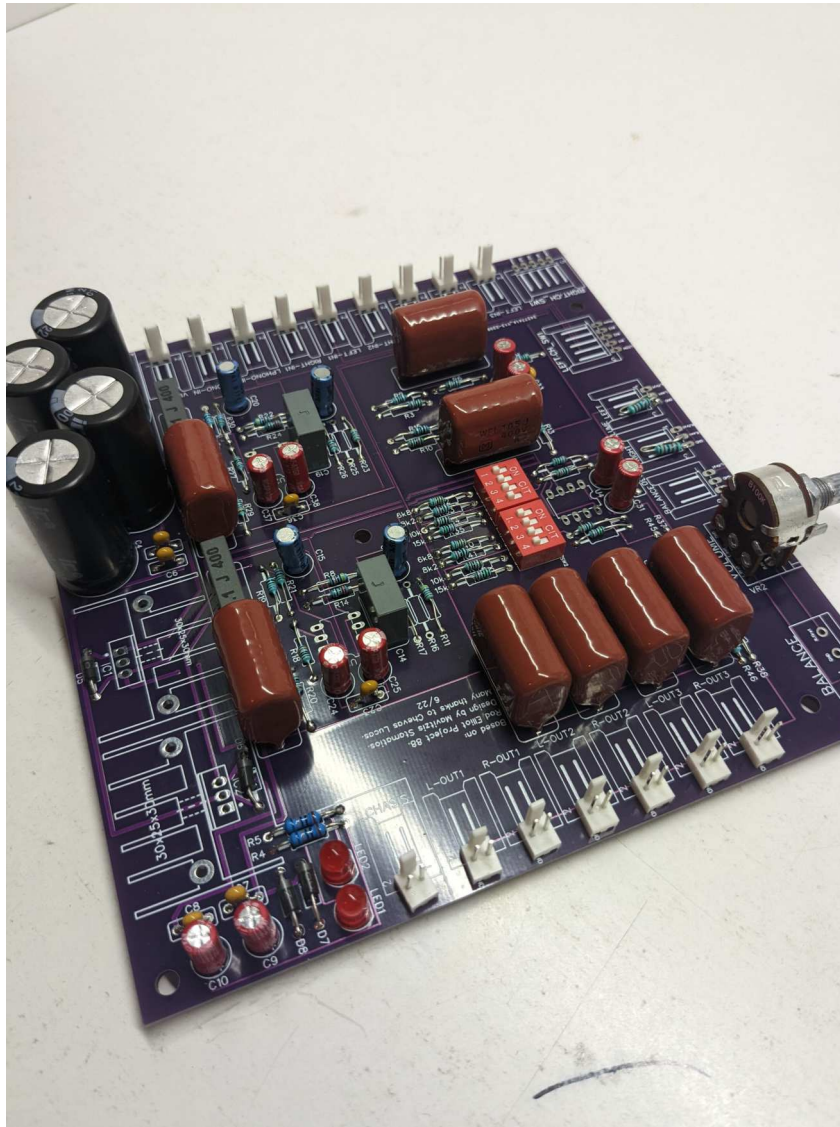


Figure B.5: Assembled main preamplifier PCB.

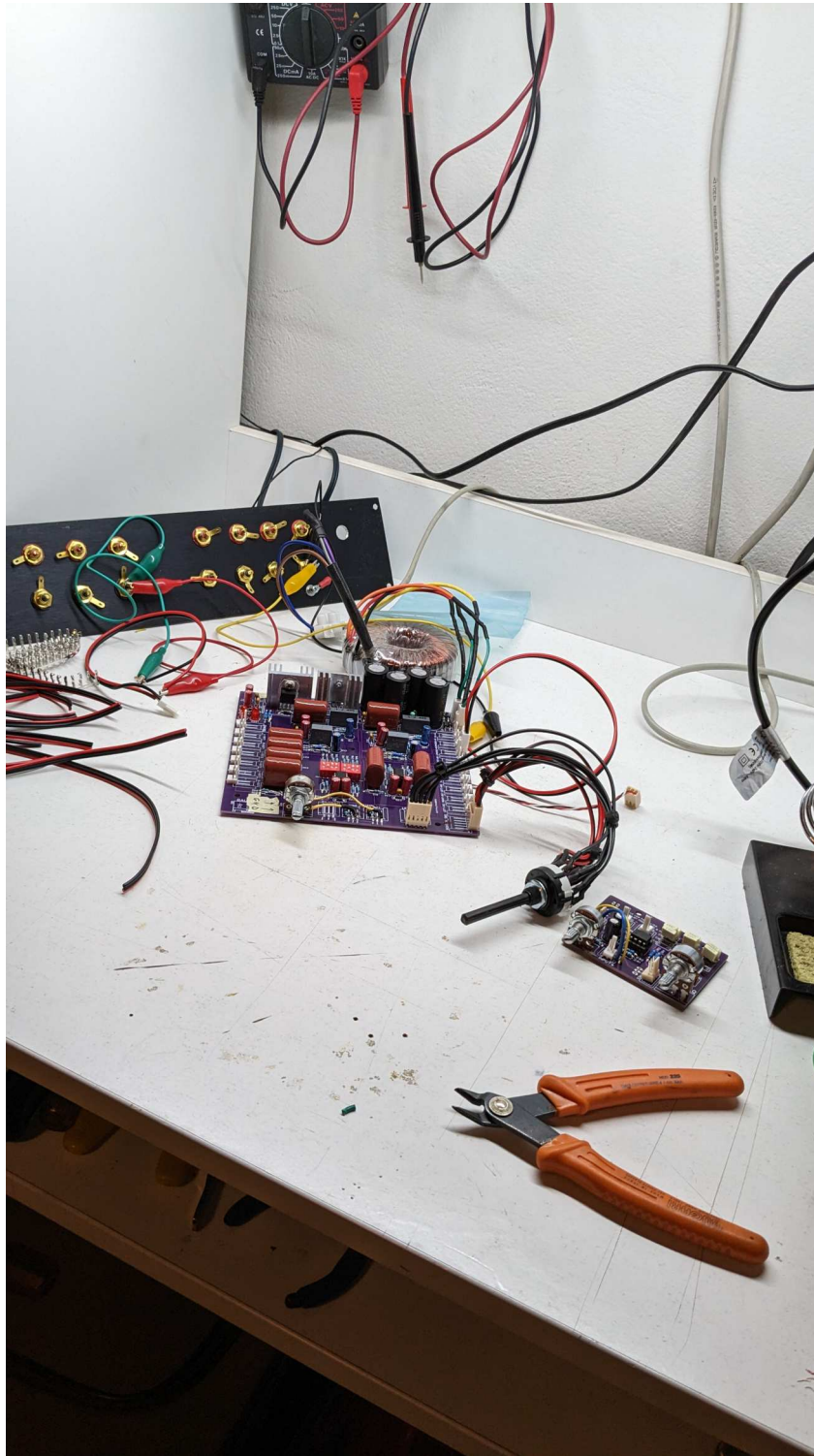


Figure B.6: Preamplifier boards and panel wiring during construction and interconnection.

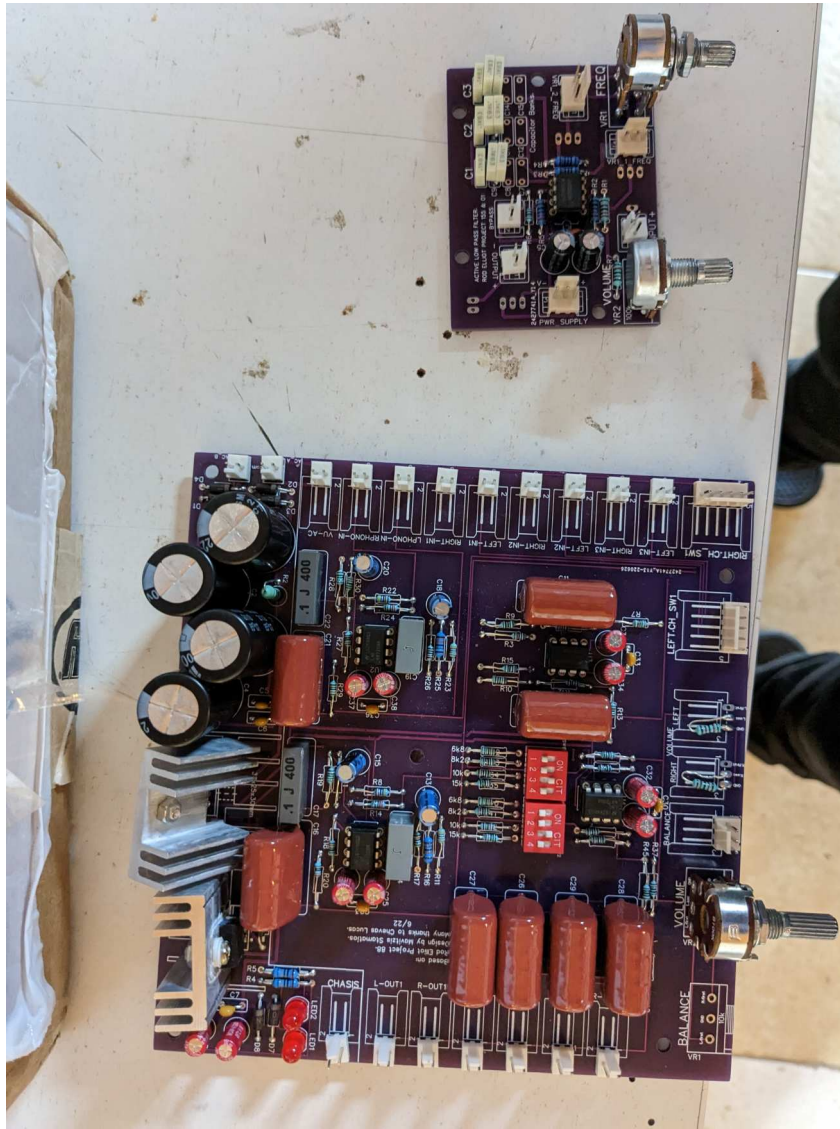


Figure B.7: Main preamplifier PCB and separate Project 155 low-pass-filter PCB.

B.3 Development Photographs

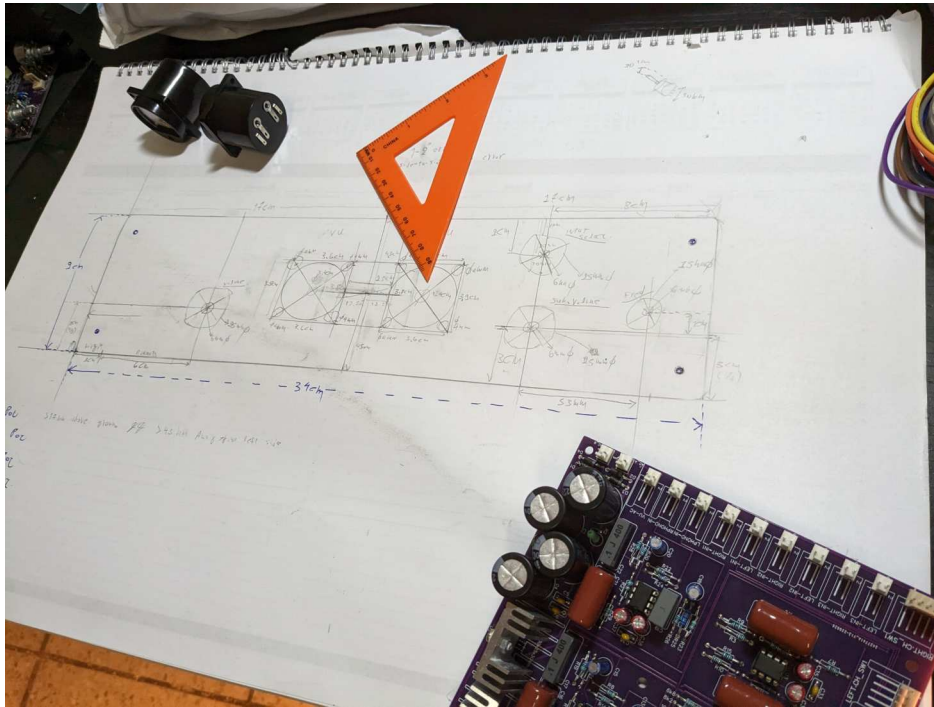


Figure B.8: Early layout and mechanical-planning work for the preamplifier.

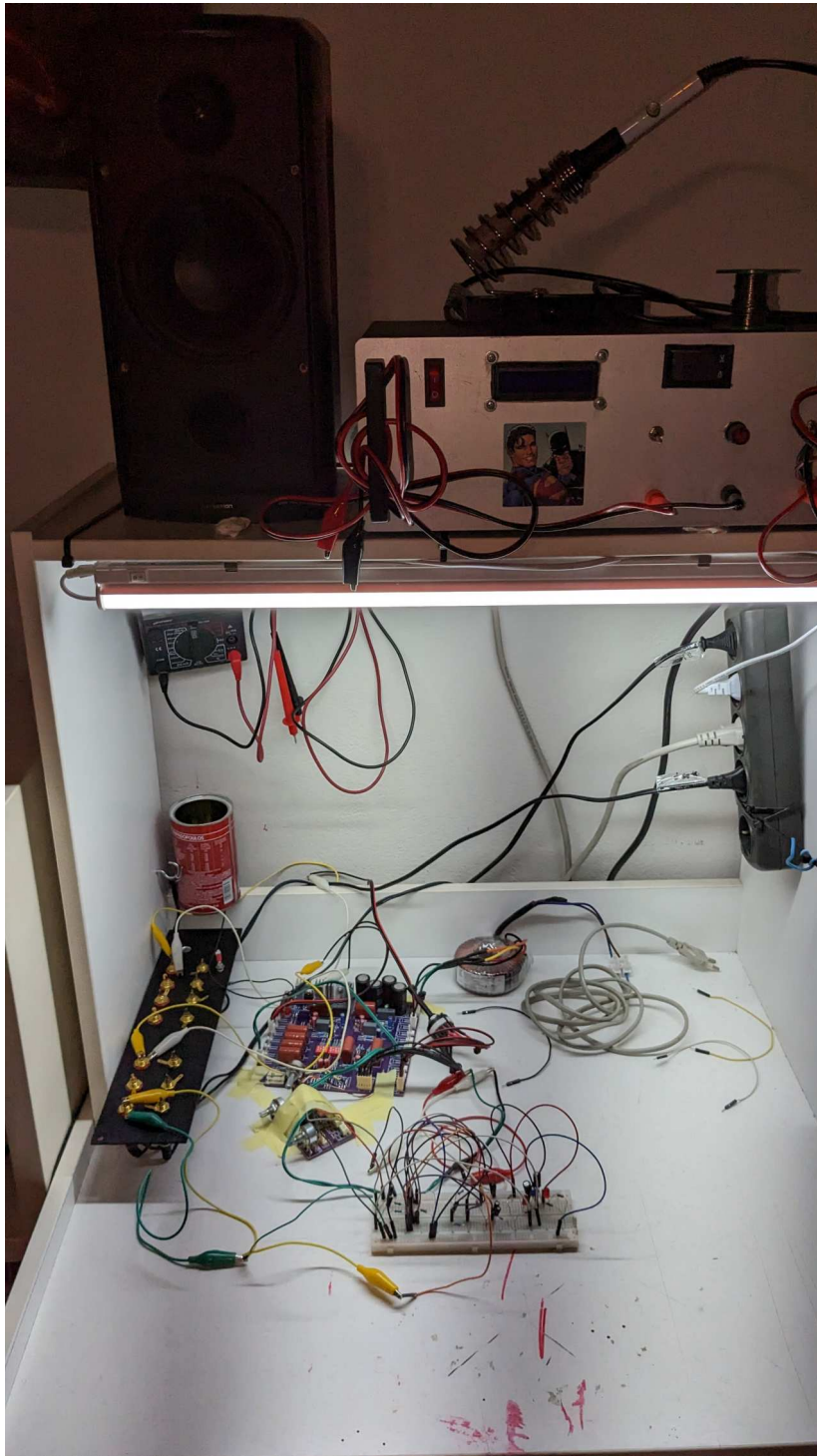


Figure B.9: Early bench assembly and wiring stage.

Appendix C

PCB Layer Artwork and Grounding Layout

This appendix presents the complete top and bottom copper-layer layouts of the main preamplifier PCB. These views provide additional documentation of the physical routing of the audio, power-supply, and ground connections.

The two layers should be considered together when examining the return-current paths. The board ground connections ultimately converge toward the common PCB ground connection, which is connected in the completed preamplifier to the dedicated chassis star-ground point. The chassis star point also provides the controlled connection between circuit ground, chassis, and protective earth.

C.1 Top Copper Layer

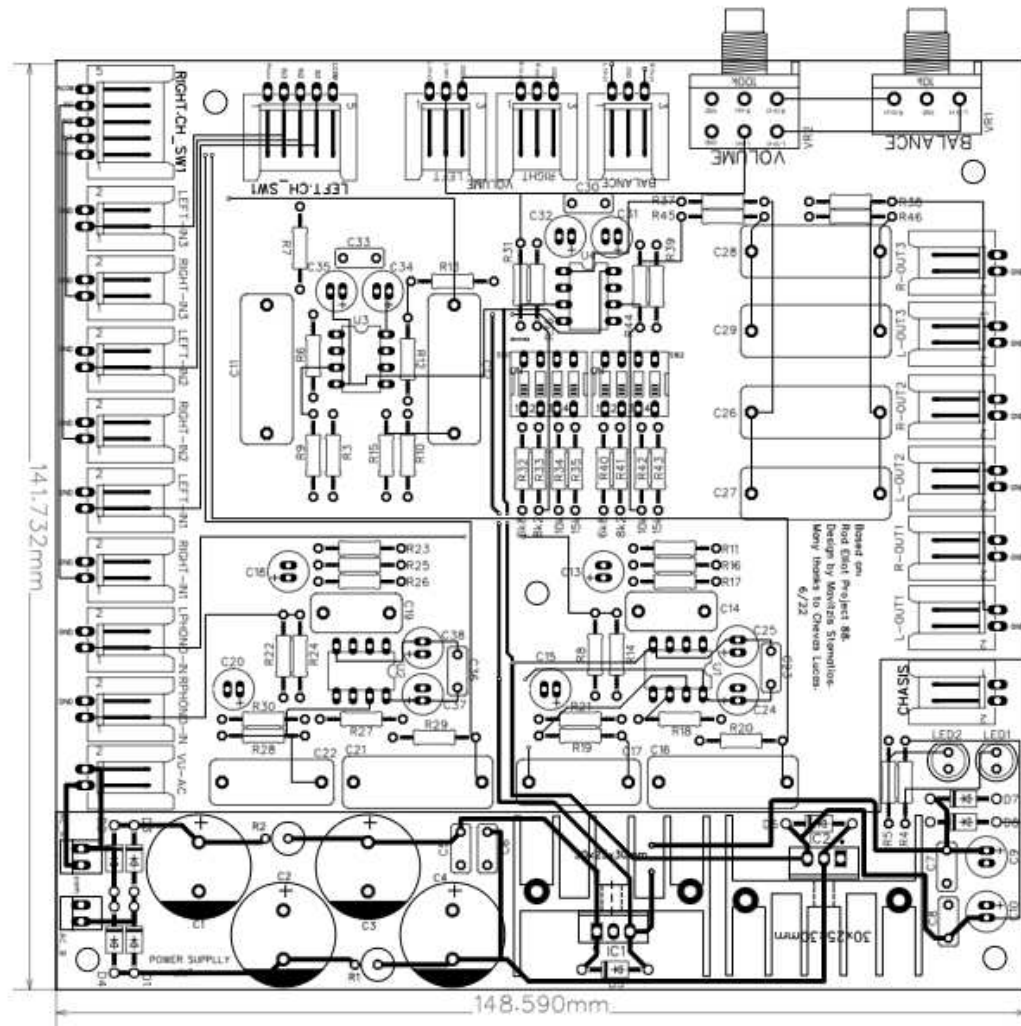


Figure C.1: Complete top copper layer of the main preamplifier PCB.

C.2 Bottom Copper Layer

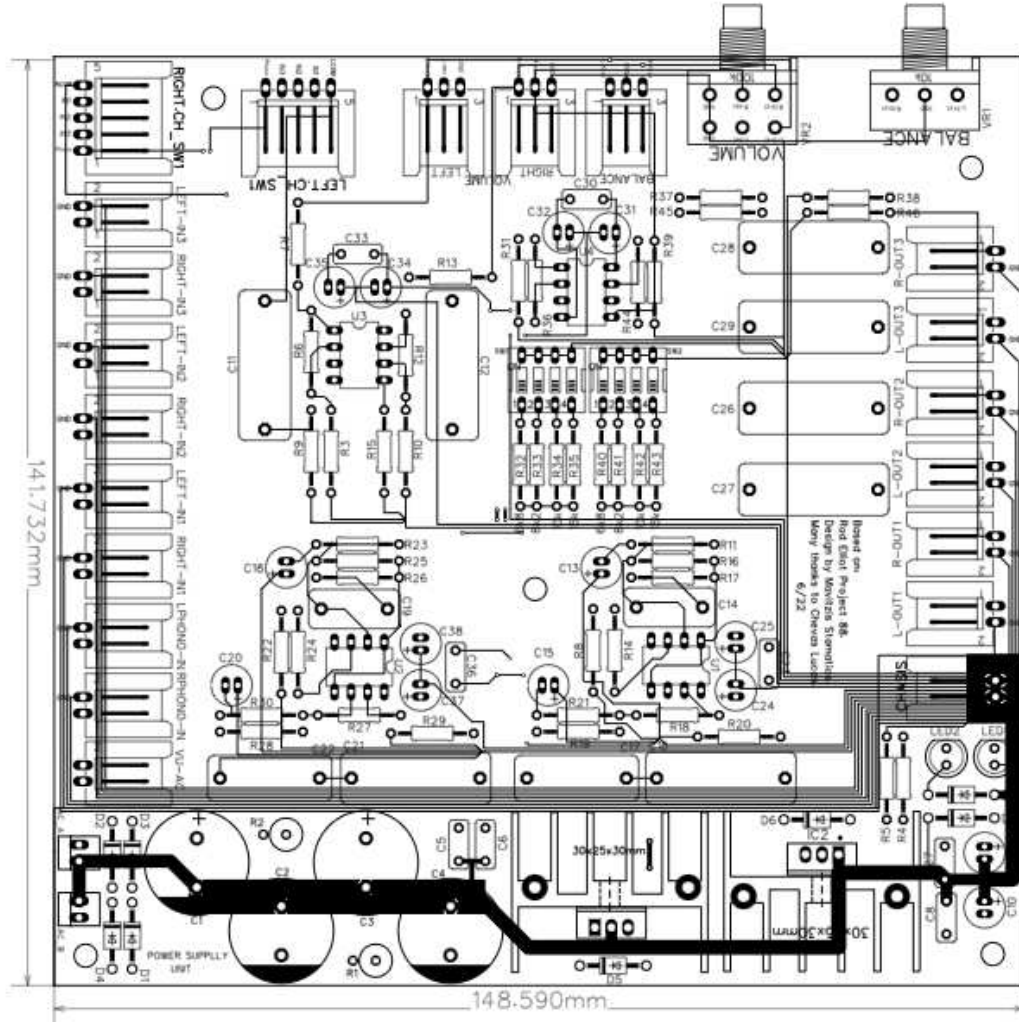


Figure C.2: Complete bottom copper layer of the main preamplifier PCB.

C.3 Ground-Path Observation

The layer views illustrate how the copper routing of the board contributes to the overall grounding strategy. The ground connections from the different functional areas of the PCB are brought toward a common board-ground connection rather than intentionally using the metal enclosure as a distributed signal-return conductor.

In the completed preamplifier, this common PCB ground is connected to the dedicated chassis star-ground point. The RCA connectors are electrically isolated from the enclosure, preventing their shells from creating additional signal-ground-to-chassis

connections. The resulting arrangement therefore provides a controlled relationship between PCB signal ground, power-supply ground, chassis, and protective earth.

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